

RESEARCH CONTINUATION

Polylogarithm Continuation

Fractional Kernels, Cayley Quotients,
and Joint Scaling Limits

Prepared for Vladimir Reshetnikov's **ProveIt** project

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Abstract

We continue the signed-kernel, level-four relation, Stieltjes–Lerch zero, and reflected log-gamma programmes in the **ProveIt** manuscript. For $H_{n-1}^{(b)}/n^a$, we classify exactly the nonnegative outer orders and positive inner orders admitting a finite signed Hausdorff moment measure. The boundary contains positive endpoint atoms. The classification yields complex nonvanishing, unique angular zeros, one-sided Euler enclosures, and power-law or exponential atom-formation scales.

We prove the sign of the first nonconstant coefficient in the existing normalized-radius conjecture for its full proposed range, including a unique fractional-order threshold. An exact interval certificate proves the emergence of a strict radial maximum below that threshold and, at another inner order, a strict radial minimum above it. For the derivative family with $n = k + r$, an angular Cauchy estimate yields a complete Bessel expansion uniform in the entire Lerch parameter interval. Its simple positive Bessel branches move only exponentially under Lerch deformation. A separate theorem gives the complete small-parameter positive-zero count for every fixed pair of indices.

For reflected log-gamma moments, the critical scale $n = (m + 1)(\log m + c)$ has a nontrivial limiting factor $\exp((\zeta(2)/(2\gamma) - \gamma)e^{-c})$; we prove the first uniform correction. Finally, the full shuffle ideal generated by convergent Cayley symmetry has an exactly determined polynomial quotient, Hilbert series, conjugation character, and coefficient asymptotics. An explicit all-index high-depth identity includes a 96-term expression for S_6 . The conjectured shorter depth-two reduction remains open.

Analytic proofs, finite rational certificates, symbolic coefficient generation, and floating-point diagnostics are identified separately. The accompanying package includes all source, reproducible verification, integration notes, and a focused programme of further research.

Source snapshot. 28357e8ca63dd78327db91d9be239d75e4462879.

Status. Ordinary mathematical proofs and reproducible exact certificates; not proof-assistant verified or externally refereed. Claims of newness refer to the inspected repository and reports.

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1 Scope, source snapshot, and result map

The source manuscript is Vladimir Reshetnikov’s **ProveIt** project, under **Analysis/Polylogarithm s/docs/manuscript**. We pinned the working branch to commit `28357e8ca63dd78327db91d9be239d75e4462879` and inspected its chapter sources, editorial ledger, and the relevant pending research reports [1, 2, 3, 4]. This matters mathematically: several older reports call S_4 a conjecture, whereas the canonical manuscript now contains an exact proof. Its specified level-four product-matrix rank conjecture is also already proved. The S_6 depth-two identity is still conjectural, and an earlier frozen S_8 search vector has already been rigorously refuted. None of those completed results is counted here as a new theorem.

The present work concentrates on places where the existing hypotheses leave a real mathematical boundary: outer orders below one, simultaneous growth of asymptotic indices, and the difference between a set of linear symmetry rows and its multiplicative closure. These boundaries lead to the principal results in Table 1.

Result	Precise advance
Sharp finite-measure domain	A necessary and sufficient condition for $H_{n-1}^{(b)}/n^a$ to have a finite signed moment measure; explicit interior densities and critical atoms (Theorem 3.1).
Complex and angular zeros	Nonvanishing on the principal slit plane and one simple imaginary-part zero on each open upper arc throughout the full measure domain (Theorem 4.1).
Endpoint scales	Two power laws, an exponential logarithmic corner, weak limits, and a sharp divergent signed-mass scale (Theorem 5.1 and Proposition 5.2).
Existing radial conjecture	The exact sign of its quadratic radial coefficient for every real $a \geq 2, b > 0$, its $a = 1$ boundary, and a unique fractional threshold (Theorem 7.2); proved local maxima and minima at two certified inner orders (Theorems 8.1 and 8.2).
Adjacent-index zeros	A complete Bessel expansion for $n = k + r$, every fixed integer r , uniform in $0 \leq \rho \leq 1$, with explicit zero coefficients and exponential deformation bounds (Theorems 9.2–9.3).
Small Lerch parameter	Complete positive-zero counts and central Puiseux branches for every fixed (n, k) (Theorem 10.1).
Growing reflected moments	A uniform transition at $n = (m + 1)(\log m + c)$, its first correction and lattice location (Theorem 11.1).
Cayley quotient and identities	Exact all-weight raw ranks and shuffle-ideal Hilbert characters; a proved explicit family containing S_6 (Theorems 12.1–12.2 and Proposition 12.4).

Table 1: Theorem-level advances relative to the inspected source. Finite computations audit these results; their all-parameter statements are established by the written proofs.

1.1 Relation to established mathematics

The regularized Hurwitz and digamma integrals used below are classical [5, 6]. The Bessel limit is proved here directly from a finite elementary-symmetric product; the Bessel series and simple positive zeros are classical [8]. Its form is reminiscent of Mehler–Heine limits, but no identification of these coefficient polynomials with a classical orthogonal family is assumed. The shuffle polynomial theorem is due to the classical work of Radford and Melançon–Reutenauer [11, 12]. The Cayley transformation itself is inherited from the manuscript’s convergent octahedral argument; it is not

introduced here as a new change of variables. The broader use of octahedral symmetry at level four has a precedent in Zhao [13].

Reflected products of powers of $\log \Gamma$ were studied by Amdeberhan, Coffey, Espinosa, Koutschan, Manna and Moll [10]. In particular, the moments, their reflection symmetry, and elementary identities obtained from Euler's reflection formula are not new. Our contribution is the joint growth regime and its controlled correction. Targeted primary-source research supports these attributions; it does not establish exhaustive historical priority for every new deduction in this article.

2 Conventions and inherited analytic framework

For real $b > 0$, let

$$H_N^{(b)} = \sum_{j=1}^N j^{-b}, \quad H_0^{(b)} = 0, \quad H_N = H_N^{(1)}.$$

We use decreasing series indices and the outer-first convention

$$\text{Li}_{s_1, \dots, s_d}(z_1, \dots, z_d) = \sum_{n_1 > \dots > n_d \geq 1} \frac{z_1^{n_1} \dots z_d^{n_d}}{n_1^{s_1} \dots n_d^{s_d}}$$

where convergent, followed by the stated analytic continuation. In particular,

$$F_{a,b}(z) = \text{Li}_{a,b}(z, 1) = \sum_{n \geq 1} \frac{H_{n-1}^{(b)}}{n^a} z^n, \quad |z| < 1.$$

The slit plane is $\mathbb{C} \setminus [1, \infty)$, and logarithms there are continued from the disk. Later, $g_{a,b} = \Im F_{a,b}(i)$ denotes a Gaussian double, $\beta(s) = \sum_{j \geq 0} (-1)^j / (2j+1)^s$ is the Dirichlet beta function, and

$$S_p = \sum_{j \geq 0} \frac{(-1)^j H_j}{(2j+1)^p}.$$

The last sum is absolute for integers $p > 1$ and converges conditionally for $p = 1$. Throughout, γ without an index is Euler's constant.

The Stieltjes constants use

$$\zeta(s, a) = \frac{1}{s-1} + \sum_{n \geq 0} \frac{(-1)^n \gamma_n(a)}{n!} (s-1)^n.$$

A superscript (k) on γ_n means differentiation in a . For $0 \leq \rho < 1$ set

$$\ell_n(\rho, a) = \sum_{j \geq 0} \frac{\rho^j \log^n(a+j)}{a+j}, \quad \mathcal{D}_{n,k}^\rho(a) = \partial_a^k \ell_n(\rho, a), \quad k \geq 1,$$

and define $\mathcal{D}_{n,k}^1(a) = \gamma_n^{(k)}(a)$. Only the differentiated family is unified at $\rho = 1$; the series defining ℓ_n generally diverges there. The sign-normalized family is

$$F_{n,k}^\rho(a) = (-1)^{n+k} \mathcal{D}_{n,k}^\rho(a).$$

The same symbol F with different indices is conventional in the source: $F_{a,b}(z)$ is a double polylogarithm, while $F_{n,k}^\rho(a)$ is the Stieltjes–Lerch derivative family. The radius ρ in the angular sections and the Lerch parameter ρ in the zero sections are separate variables.

The master coefficient formula, reproduced in (9.1), follows from differentiating $(a + j)^{-s}$ in a and extracting the coefficient of $s - 1$. For $k \geq 1$ its shift series and every fixed number of a derivatives converge locally uniformly for $a > 0$, uniformly in $0 \leq \rho \leq 1$. The bound is a fixed power of $\log(a + j)$ divided by $(a + j)^{k+1}$. This convergence justifies the classical endpoint without assigning a value to the divergent undifferentiated series.

All finite signed measures in the article are real measures of finite total variation. Weak convergence means convergence against every continuous function on $[0, 1]$. Big- O constants may depend on fixed indices and fixed compact parameter sets; every claimed uniformity is stated explicitly.

3 The sharp signed-moment threshold for subunit orders

The earlier angular-zero continuation [2] extends the signed Mellin kernel from integer outer order to real $a \geq 1$. The restriction $a \geq 1$ is not the actual boundary of this method. The correct boundary is $a + b = 1$ when $0 < b < 1$. At that boundary a density becomes a measure with a positive atom at one. The atom must be retained: substituting zero for a vanishing reciprocal Gamma factor would lose its entire mass.

Write

$$A_n(a, b) = \frac{H_{n-1}^{(b)}}{n^a}, \quad F_{a,b}(z) = \sum_{n \geq 1} A_n(a, b) z^n \quad (|z| < 1).$$

We permit $a = 0$ to state the sharp endpoint. Throughout $b > 0$.

Theorem 3.1 (Exact finite-measure domain). *For $a \geq 0$ and $b > 0$, a finite real signed measure $\nu_{a,b}$ on $[0, 1]$ satisfying*

$$\int_{[0,1]} u^{n-1} d\nu_{a,b}(u) = A_n(a, b) \quad (n \geq 1) \quad (3.1)$$

exists if and only if

$$(a > 0 \text{ and } a + b \geq 1) \quad \text{or} \quad (a = 0 \text{ and } b > 1). \quad (3.2)$$

It is unique and has total mass zero. More precisely:

1. *If $a > 0$ and $a + b > 1$, it has an integrable density $\kappa_{a,b}(u)$ with exactly one strict sign change, from negative to positive as u increases, at a point $r_{a,b} \in (0, 1)$.*
2. *If $0 < b < 1$ and $a = 1 - b$, then $\nu_{a,b} = (1 - b)^{-1} \delta_1 + \kappa_{a,b}(u) du$, where $\kappa_{a,b}(u) < 0$ throughout $(0, 1)$.*
3. *If $a = 0$ and $b > 1$, then $\nu_{0,b} = \zeta(b) \delta_1 + \kappa_{0,b}(u) du$, where $\kappa_{0,b}(u) < 0$ throughout $(0, 1)$.*

In the two atomic cases all positive mass is at one and all negative mass is in $(0, 1)$. Outside (3.2), not even a finite signed measure with arbitrarily many sign changes can satisfy (3.1).

3.1 A regularized positive kernel

Set

$$q(t) = \frac{1}{1 - e^{-t}} - \frac{1}{t}, \quad t > 0. \quad (3.3)$$

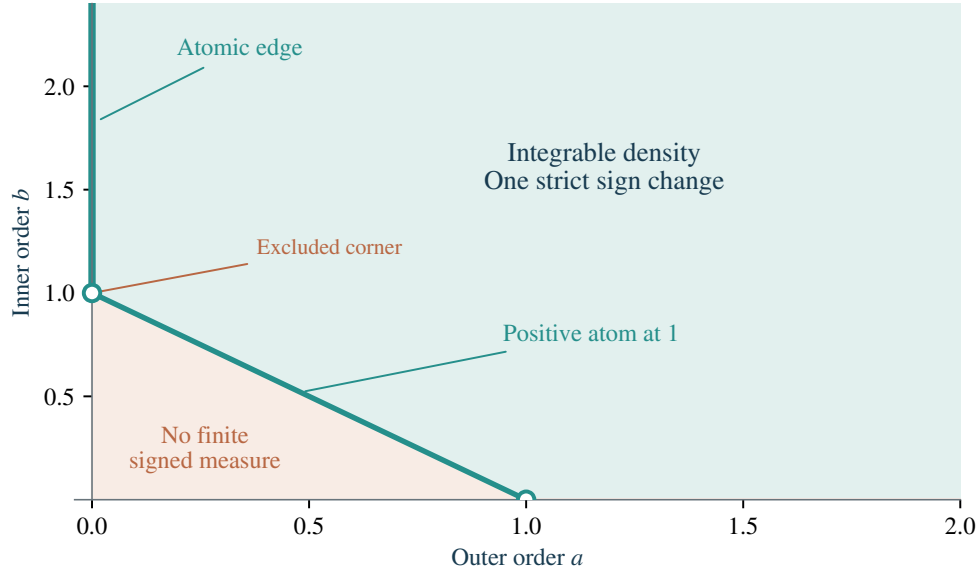


Figure 1: The exact finite signed-measure domain for $a \geq 0, b > 0$. The teal edges carry a positive atom at one. The point $(0, 1)$ is excluded, and the horizontal axis $b = 0$ is outside the stated domain. The absence of a finite measure below the diagonal does not assert the presence of a complex zero.

The elementary identities

$$q(t) = \frac{1}{2} + \frac{1}{2} \coth(t/2) - \frac{1}{t}, \quad q'(t) = \frac{1}{t^2} - \frac{1}{4 \sinh^2(t/2)} > 0$$

give $1/2 < q(t) < 1$, because $\sinh(t/2) > t/2$ and the endpoint limits are $1/2$ and 1 . In particular q is positive, bounded and strictly increasing. Also $q(t) = 1/2 + O(t)$ as $t \downarrow 0$.

The classical Hurwitz integral and one elementary subtraction give

$$\zeta(b, x) = \frac{x^{1-b}}{b-1} + \frac{1}{\Gamma(b)} \int_0^\infty e^{-xt} t^{b-1} q(t) dt, \quad x > 0, b > 0, b \neq 1. \quad (3.4)$$

For $b > 1$ this is obtained by splitting $1/(1 - e^{-t}) = 1/t + q(t)$ in the usual Gamma integral. The integral involving q is holomorphic for $\Re b > 0$, so the identity theorem extends the same formula to $0 < b < 1$. This is analytic continuation of an absolutely convergent regularized integral, not subtraction of two divergent real integrals. The standard starting integral and recurrence are DLMF 25.11.25 and 25.11.4 [5]. For $b = 1$ the corresponding formula is

$$\psi(x) = \log x - \int_0^\infty e^{-xt} q(t) dt, \quad (3.5)$$

which is DLMF 5.9.13 [6] and also follows by taking the finite part of (3.4) at $b = 1$.

For $0 < b < 1$, we shall use $\zeta(b) < 0$. Indeed the alternating Dirichlet series $\eta(b)$ is positive, while $\eta(b) = (1 - 2^{1-b})\zeta(b)$ and $1 - 2^{1-b} < 0$.

3.2 Explicit densities and their Laplace transforms

Put $T = -\log u$ and $K_{a,b}(T) = \kappa_{a,b}(e^{-T})$. For $a > 0$, $a + b > 1$ and $b \neq 1$, define

$$K_{a,b}(T) = \frac{\zeta(b)T^{a-1}}{\Gamma(a)} - \frac{T^{a+b-2}}{(b-1)\Gamma(a+b-1)} - \frac{1}{\Gamma(a)\Gamma(b)} \int_0^T (T-t)^{a-1} t^{b-1} q(t) dt. \quad (3.6)$$

When $b > 1$, an equivalent form is

$$K_{a,b}(T) = \frac{\zeta(b)T^{a-1}}{\Gamma(a)} - \frac{1}{\Gamma(a)\Gamma(b)} \int_0^T \frac{(T-t)^{a-1} t^{b-1}}{1-e^{-t}} dt. \quad (3.7)$$

At $b = 1$, define

$$K_{a,1}(T) = \frac{T^{a-1}}{\Gamma(a)} (\gamma + \psi(a) - \log T) - \frac{1}{\Gamma(a)} \int_0^T (T-t)^{a-1} q(t) dt. \quad (3.8)$$

The Gamma factors and the term $\gamma + \psi(a)$ in this last formula are essential; in particular it reduces at $a = 1$ to $K_{1,1}(T) = -\log(e^T - 1)$, agreeing with the original signed kernel.

These functions are integrable against $e^{-T} dT$. At zero, (3.6) has only the integrable powers T^{a-1} , T^{a+b-2} and $O(T^{a+b-1})$; the logarithmic factor in (3.8) is likewise integrable for $a > 0$. At infinity all terms have at most polynomial times logarithmic growth. For $b > 1$, the integrability of the convolution in (3.7) at its lower endpoint uses $b > 1$ explicitly.

By the Gamma integral and the convolution rule, the Laplace transform of (3.6) is

$$\begin{aligned} \int_0^\infty e^{-nT} K_{a,b}(T) dT &= n^{-a} \left[\zeta(b) - \frac{n^{1-b}}{b-1} - \frac{1}{\Gamma(b)} \int_0^\infty e^{-nt} t^{b-1} q(t) dt \right] \\ &= n^{-a} (\zeta(b) - \zeta(b, n)) = \frac{H_{n-1}^{(b)}}{n^a}. \end{aligned}$$

Each convolution interchange is justified absolutely; q is bounded and the remaining factors have Gamma integrals. Formula (3.8) has Laplace transform

$$n^{-a} \left(\gamma + \log n - \int_0^\infty e^{-nt} q(t) dt \right) = n^{-a} (\gamma + \psi(n)) = \frac{H_{n-1}}{n^a}.$$

Here the Laplace transform of $T^{a-1} \log T$ is obtained by differentiating the Gamma integral, justified by absolute integrability. The substitution $u = e^{-T}$ changes $u^{n-1} du$ to $e^{-nT} dT$, proving the desired moment normalization. In particular $n = 1$ gives total mass zero.

3.3 The one-sign-change proof

For $b > 1$, divide (3.7) by T^{a-1} and rescale $t = Tv$:

$$T^{1-a} K_{a,b}(T) = \frac{\zeta(b)}{\Gamma(a)} - \frac{1}{\Gamma(a)\Gamma(b)} \int_0^1 (1-v)^{a-1} v^{b-1} \frac{T^b}{1-e^{-Tv}} dv.$$

For fixed $v > 0$, the logarithmic derivative of $T^b/(1 - e^{-Tv})$ is

$$\frac{b}{T} - \frac{v}{e^{Tv} - 1} > \frac{b-1}{T} > 0.$$

Thus the rescaled density is strictly decreasing. Its limit at zero is $\zeta(b)/\Gamma(a) > 0$, and its limit at infinity is $-\infty$. This proves one strict positive–negative crossing in the T coordinate.

For $b = 1$, the rescaled expression is

$$\Gamma(a)T^{1-a}K_{a,1}(T) = \gamma + \psi(a) - \log T - T \int_0^1 (1-v)^{a-1} q(Tv) dv.$$

It is strictly decreasing from $+\infty$ to $-\infty$, since q is positive and increasing. For $0 < b < 1$, instead divide by T^{a+b-2} to obtain

$$\begin{aligned} T^{2-a-b}K_{a,b}(T) &= \frac{1}{(1-b)\Gamma(a+b-1)} + \frac{\zeta(b)}{\Gamma(a)}T^{1-b} \\ &\quad - \frac{T}{\Gamma(a)\Gamma(b)} \int_0^1 (1-v)^{a-1}v^{b-1}q(Tv) dv. \end{aligned} \quad (3.9)$$

Now the first term is positive, the second is strictly decreasing because $\zeta(b) < 0$, and the final term is strictly decreasing because q is positive and increasing. The endpoint limits are a positive constant and $-\infty$. Thus there is again exactly one strict crossing. The derivatives of these rescaled expressions are strictly negative; differentiation is dominated by the same integrable beta weights. At a zero of K , multiplication by the positive rescaling factor does not affect the sign of its derivative. Reversing $T = -\log u$ gives the asserted negative–positive sign change and a strictly positive u derivative at its unique crossing.

3.4 The critical atom and the exact boundary

Let $0 < b < 1$ and $a = 1 - b$. Replace the middle term of (3.6) by an atom of mass $(1 - b)^{-1}$ at $u = 1$, and retain the density

$$K_{1-b,b}(T) = \frac{\zeta(b)T^{-b}}{\Gamma(1-b)} - \frac{1}{\Gamma(1-b)\Gamma(b)} \int_0^T (T-t)^{-b}t^{b-1}q(t) dt. \quad (3.10)$$

Both terms are strictly negative and integrable against $e^{-T} dT$. Its moments plus the atom are

$$\frac{1}{1-b} + n^{b-1} \left[\zeta(b) - \frac{1}{\Gamma(b)} \int_0^\infty e^{-nt}t^{b-1}q(t) dt \right] = \frac{H_{n-1}^{(b)}}{n^{1-b}},$$

by (3.4). Thus the measure has total mass zero, and its negative mass is exactly $(1 - b)^{-1}$.

For $a = 0$, $b > 1$, the formula is especially simple:

$$\nu_{0,b} = \zeta(b)\delta_1 + K_{0,b}(-\log u) du, \quad K_{0,b}(T) = -\frac{T^{b-1}}{\Gamma(b)(1 - e^{-T})}. \quad (3.11)$$

The ordinary Hurwitz integral gives its moments immediately. Its negative mass is exactly $\zeta(b)$.

Every finite signed measure has bounded moments, since $|\int u^{n-1} d\nu| \leq \|\nu\|_{TV}$. If $0 < b < 1$, elementary integral comparison gives $H_{n-1}^{(b)} \sim n^{1-b}/(1-b)$, so the moments diverge to infinity when $a + b < 1$. If $a = 0, b = 1$, they diverge logarithmically. These are exactly the excluded cases for $a \geq 0, b > 0$. Finally, two finite signed measures with the same moments integrate every polynomial equally; polynomial density in $C([0, 1])$ makes the measures identical. This proves Theorem 3.1.

4 Consequences for complex zeros and angular zeros

Theorem 4.1 (Nonvanishing throughout the finite-measure domain). *For all parameters in (3.2), $F_{a,b}$ has an analytic continuation to $\mathbb{C} \setminus [1, \infty)$ and has no zero there except for its double zero at the origin. Every upper open semicircle $z = \rho e^{i\theta}$, $0 < \rho \leq 1$, $0 < \theta < \pi$, has exactly one simple zero of $\text{Im } F_{a,b}(z)$, and it satisfies*

$$\arccos \rho < \theta_{a,b}(\rho) < \frac{\pi}{2}.$$

The imaginary part is positive below this angle and negative above it; its angular derivative at the zero is strictly negative. In the atomic cases, unit-circle values mean analytic continuation or Abel boundary values: the original Fourier summands do not tend to zero, so ordinary Fourier convergence is not asserted.

Proof. The signed transform

$$F_{a,b}(z) = z \int_{[0,1]} \frac{d\nu_{a,b}(u)}{1 - zu}$$

follows by geometric expansion in the disk and extends analytically to the slit plane by uniform domination on compact sets. Let r be the crossing from Theorem 3.1, or let $r = 1$ in the atomic cases. Zero mass gives

$$F_{a,b}(z) = \frac{z^2}{1 - rz} P_{a,b}(z), \quad P_{a,b}(z) = \int_{[0,1]} \frac{u - r}{1 - zu} d\nu_{a,b}(u). \quad (4.1)$$

The measure $(u - r) d\nu_{a,b}(u)$ is positive and nonzero, including when $r = 1$: its atom disappears but its density is strictly positive on $(0, 1)$. Consequently $\text{Im } P_{a,b}(z)$ has the strict sign of $\text{Im } z$, while $P_{a,b}(x) > 0$ for $x < 1$. Apart from the explicit z^2 , the factor $(1 - rz)^{-1}$ in (4.1) is analytic and nonzero on the slit plane. Since $P_{a,b}(0) = A_2 = 2^{-a}$, the zero at zero has multiplicity exactly two.

For angular zeros put

$$J_\rho(c) = \int_{[0,1]} \frac{d\nu(u)}{1 - 2\rho cu + \rho^2 u^2}, \quad \text{Im } F(\rho e^{i\theta}) = \rho \sin \theta J_\rho(\cos \theta).$$

The zero-mass one-crossing sign test holds for our atomic measures as well: subtract the value of the test function at r , and use the ordered supports of the two signs. It gives $J_\rho(0) < 0$. When $\rho < 1$, the kernel at $c = \rho$ is strictly increasing in u , so $J_\rho(\rho) > 0$. When $\rho = 1$ and the crossing is interior, separate the two signs before expanding $(1 - u)^{-2}$ to obtain

$$\lim_{c \uparrow 1} J_1(c) = \int_{[0,1]} \frac{d\nu(u)}{(1 - u)^2} = \sum_{n \geq 1} n A_n > 0,$$

allowing $+\infty$. The negative part is supported away from one, so this is the same justified monotone-limit argument as in the original signed-kernel theorem. In the atomic cases the limiting signed integral at $c = 1$ need not be defined, and we do not use that display. Instead the limit is $+\infty$ directly: the atom contributes $M/[2(1 - c)]$, and the negative density contributes $o((1 - c)^{-1})$. For the latter assertion, multiply its integral by $1 - c$ and use dominated convergence: the multiplier $(1 - c)/(1 - 2cu + u^2)$ is bounded by 1 for $0 \leq c < 1$ and tends to zero for every $u < 1$. The negative density is integrable and has no atom at one.

For $c' > c$, the quotient of the two kernels has derivative

$$\frac{d}{du} \frac{1 - 2\rho cu + \rho^2 u^2}{1 - 2\rho c' u + \rho^2 u^2} = \frac{2\rho(c' - c)(1 - \rho^2 u^2)}{(1 - 2\rho c' u + \rho^2 u^2)^2} > 0 \quad (0 < u < 1).$$

At any zero, apply the sign test to the signed measure obtained by weighting ν with the positive kernel. It follows that J_ρ is positive at every larger c and negative at every smaller c ; this proves uniqueness and the signs. Finally its derivative at its zero is positive, because the additional factor $2\rho u/(1 - 2\rho cu + \rho^2 u^2)$ is strictly increasing in u . The angular derivative is therefore $-\rho \sin^2 \theta J'_\rho(\cos \theta) < 0$. $\square$

The theorem is sharp as a statement about finite signed moment representations. It does not assert that nonvanishing fails below $a + b = 1$: the absence of this particular representation is not a complex-zero counterexample. In fact the entire axis $a = 0, b > 0$ is zero-free away from the origin. The geometric sum gives

$$F_{0,b}(z) = \frac{z \operatorname{Li}_b(z)}{1 - z} = \frac{z^2}{1 - z} \frac{1}{\Gamma(b)} \int_0^1 \frac{(-\log u)^{b-1}}{1 - zu} du.$$

The last integral is a nonzero positive-measure transform on the slit plane, by exactly the imaginary-part and real-axis argument used above. Thus finite signed-moment representability and complex nonvanishing have different parameter boundaries, already on this explicit axis.

5 Formation of the endpoint atom

Theorem 5.1 (Boundary-layer scale and weak convergence). *Fix $0 < b < 1$, let $a = 1 - b + \varepsilon$ with $\varepsilon > 0$, and let $T_\varepsilon = -\log r_{a,b}$. Define*

$$\tau_\varepsilon = \left(\frac{\Gamma(1 - b + \varepsilon)}{(1 - b)(-\zeta(b))\Gamma(\varepsilon)} \right)^{1/(1-b)}.$$

Then, as $\varepsilon \downarrow 0$,

$$T_\varepsilon = \tau_\varepsilon \left(1 + O_b \left(\frac{\tau_\varepsilon}{\varepsilon} \right) \right), \quad 0 < T_\varepsilon < \tau_\varepsilon, \quad (5.1)$$

and in particular

$$1 - r_{1-b+\varepsilon,b} \sim \left(\frac{\Gamma(1 - b)}{(1 - b)(-\zeta(b))\varepsilon} \right)^{1/(1-b)}. \quad (5.2)$$

The measures converge weakly to the critical measure in Theorem 3.1, and their positive masses converge to $(1 - b)^{-1}$ while their positive supports shrink to $u = 1$.

Proof. In (3.9), put $A_\varepsilon = ((1 - b)\Gamma(\varepsilon))^{-1}$ and $B_\varepsilon = (-\zeta(b))/\Gamma(1 - b + \varepsilon)$. The zero equation is

$$A_\varepsilon - B_\varepsilon T^{1-b} - TC_\varepsilon(T) = 0, \quad C_\varepsilon(T) = \frac{1}{\Gamma(a)\Gamma(b)} \int_0^1 (1 - v)^{a-1} v^{b-1} q(Tv) dv.$$

Since $1/2 < q < 1$ and $a + b = 1 + \varepsilon$,

$$\frac{1}{2\Gamma(1 + \varepsilon)} < C_\varepsilon(T) < \frac{1}{\Gamma(1 + \varepsilon)}.$$

By definition $B_\varepsilon \tau_\varepsilon^{1-b} = A_\varepsilon$, so the equation at $T = \tau_\varepsilon$ has strictly negative left side and $T_\varepsilon < \tau_\varepsilon$. Furthermore

$$0 \leq 1 - (T_\varepsilon/\tau_\varepsilon)^{1-b} = \frac{T_\varepsilon C_\varepsilon(T_\varepsilon)}{A_\varepsilon} \leq \frac{(1 - b)\tau_\varepsilon}{\varepsilon}.$$

The final equality in the upper bound uses $\Gamma(1 + \varepsilon) = \varepsilon\Gamma(\varepsilon)$. Now $\tau_\varepsilon = O_b(\varepsilon^{1/(1-b)})$, hence $\tau_\varepsilon/\varepsilon \rightarrow 0$. The last inequality proves (5.1). Expanding the Gamma ratio and using $1 - e^{-T} \sim T$ proves (5.2).

For weak convergence, the positive term in (3.6), regarded as a measure in the T coordinate, is

$$\frac{1}{1-b} \frac{e^{-T} T^{\varepsilon-1}}{\Gamma(\varepsilon)} dT.$$

It is $(1-b)^{-1}$ times a $\text{Gamma}(\varepsilon, 1)$ law. Its mean is ε , so it converges weakly to mass $(1-b)^{-1}$ at $T = 0$, equivalently at $u = 1$. The two negative components in (3.6) converge in $L^1(e^{-T} dT)$ to those in (3.10). For the power term, dominated convergence holds with an exponent bounded above -1 . For the convolution, either use the rescaled beta integral and the same domination, or use L^1 continuity of the Gamma densities and the integrable convolution factor $e^{-t} t^{b-1} q(t)$.

The unique positive support is $0 < T < T_\varepsilon$, which shrinks to zero. The $\text{Gamma}(\varepsilon, 1)$ mass in this interval tends to one: since T_ε is a fixed power of ε to leading order,

$$\frac{1}{\Gamma(\varepsilon)} \int_0^{T_\varepsilon} e^{-T} T^{\varepsilon-1} dT = \frac{T_\varepsilon^\varepsilon}{\Gamma(1+\varepsilon)} (1 + o(1)) \rightarrow 1.$$

The negative components have vanishing mass on the shrinking interval by their L^1 convergence to an integrable limit. Thus the actual positive mass of the signed density tends to $(1-b)^{-1}$, completing the proof. $\square$

5.1 The other edge and the logarithmic corner

The second atomic edge has a complementary power law. The corner $b = 1$ has an exponential scale and no finite limiting measure.

Proposition 5.2 (Two further endpoint scales). *For fixed $b > 1$, as $a \downarrow 0$,*

$$1 - r_{a,b} \sim (\Gamma(b)\zeta(b)a)^{1/(b-1)}. \quad (5.3)$$

The measures converge weakly to $\nu_{0,b}$, and their positive masses converge to $\zeta(b)$. For $b = 1$, set $\sigma_a = \exp(\gamma + \psi(a))$. Then

$$-\log r_{a,1} = \sigma_a(1 + O(\sigma_a/a)), \quad 0 < -\log r_{a,1} < \sigma_a, \quad (5.4)$$

where

$$\sigma_a = \exp(-1/a + \zeta(2)a - \zeta(3)a^2 + O(a^3)).$$

If $M_{a,1}$ denotes either signed mass of $\nu_{a,1}$, then

$$M_{a,1} \sim \frac{1}{ea}. \quad (5.5)$$

Thus the finite signed measures have unbounded total variation at this corner, consistently with the impossibility of a finite measure at $a = 0, b = 1$.

Proof. For $b > 1$, the equation obtained from (3.6) after division by T^{a-1} is

$$0 = \frac{\zeta(b)}{\Gamma(a)} - \frac{T^{b-1}}{(b-1)\Gamma(a+b-1)} - \frac{T^b}{\Gamma(a)\Gamma(b)} \int_0^1 (1-v)^{a-1} v^{b-1} q(Tv) dv.$$

The last integral, after its Gamma normalization, lies between $1/(2\Gamma(a+b))$ and $1/\Gamma(a+b)$. The ratio of this last term to the preceding negative term is therefore $O_b(T)$, uniformly for small positive a . The balancing point of the first two terms is

$$\left(\frac{(b-1)\Gamma(a+b-1)\zeta(b)}{\Gamma(a)} \right)^{1/(b-1)} \sim (\Gamma(b)\zeta(b)a)^{1/(b-1)}.$$

The strict monotonicity of the rescaled density, or the same two-sided comparison used in Theorem 5.1, makes the actual zero asymptotic to this point. The positive term is $\zeta(b)$ times a $\text{Gamma}(a, 1)$ law in the T coordinate. The negative term in (3.7), after multiplication by e^{-T} , is the convolution of that Gamma probability density with $e^{-t}t^{b-1}/(\Gamma(b)(1-e^{-t}))$, an integrable function of mass $\zeta(b)$. As $a \downarrow 0$, this convolution converges in L^1 to the latter function; $\text{Gamma}(a, 1)$ is an approximate identity since its mean tends to zero. This proves weak convergence. As before, the Gamma mass below the crossing tends to one, since $a \log T \rightarrow 0$, and the limiting negative density is integrable. Hence the positive mass tends to $\zeta(b)$.

For $b = 1$, put $J_a(T) = \int_0^1 (1-v)^{a-1} q(Tv) dv$, so that $1/(2a) < J_a(T) < 1/a$. At the crossing T_a , formula (3.8) says

$$\log(\sigma_a/T_a) = T_a J_a(T_a), \quad 0 < \log(\sigma_a/T_a) < \sigma_a/a.$$

This gives (5.4). The digamma expansion $\psi(a) = -1/a - \gamma + \zeta(2)a - \zeta(3)a^2 + O(a^3)$ gives the stated scale.

Finally the positive mass is the integral of $e^{-T}T^{a-1}[\log(\sigma_a/T) - TJ_a(T)]/\Gamma(a)$ over $0 < T < T_a$. Omitting the factor e^{-T} , the TJ_a term, and the vanishing interval between T_a and σ_a changes the result by a relative $o(1)$. To see this explicitly, the first omission has relative error at most σ_a ; the TJ_a integral is at most $\sigma_a^{a+1}/(a(a+1)\Gamma(a))$, whereas the main integral is $\sigma_a^a/(a^2\Gamma(a))$. For the removed interval, write $T = \sigma_a v$ and use $1 - T_a/\sigma_a = O(\sigma_a/a)$; its relative contribution is $O(\sigma_a^2)$. Therefore

$$M_{a,1} \sim \frac{\sigma_a^a}{a^2\Gamma(a)} \sim \frac{1}{ea},$$

as asserted. □

6 Euler certificates in the extended range

Put $f(k) = A_{2k+1}(a, b)$ and let $Df(k) = f(k) - f(k+1)$. The same signed-moment proof gives

$$D^k f(0) = \int (1-u^2)^k d\nu_{a,b}(u) < 0 \quad (k \geq 1), \quad D^0 f(0) = 0.$$

This strict sign remains valid in the atomic cases because the atom is annihilated for $k \geq 1$ and the density is negative. The geometric series is dominated by one, so finite variation and dominated convergence give, including the atomic cases,

$$\sum_{k \geq 0} \frac{D^k f(0)}{2^{k+1}} = \int \sum_{k \geq 0} \frac{(1-u^2)^k}{2^{k+1}} d\nu_{a,b}(u) = \int \frac{d\nu_{a,b}(u)}{1+u^2} = \Im F_{a,b}(i).$$

Thus the Euler approximants decrease to $g_{a,b} = \Im F_{a,b}(i)$ even where the original alternating Fourier series does not converge. Explicitly, the approximant with N terms is

$$E_N = \sum_{k=0}^{N-1} \frac{D^k f(0)}{2^{k+1}}, \quad N \geq 1.$$

For $b \neq 1$, the explicit decompositions prove

$$E_N - C_b 2^{-N} \leq g_{a,b} \leq E_N, \quad C_b = \begin{cases} (1-b)^{-1}, & 0 < b < 1, \\ \zeta(b), & b > 1. \end{cases}$$

Indeed either signed mass is at most C_b : use the positive Gamma component for $b < 1$ and the positive $\zeta(b)$ Gamma component for $b > 1$, together with zero mass. Each finite difference has absolute value at most that signed mass, since its weight lies in $[0, 1]$. For $b = 1$, a valid finite mass bound is

$$C_{a,1} = \frac{\exp(a(\gamma + \psi(a)))}{a^2 \Gamma(a)}.$$

Indeed the positive part of (3.8) is bounded by $T^{a-1}(\gamma + \psi(a) - \log T)_+/\Gamma(a)$, and its unweighted T integral is exactly $C_{a,1}$; the factor e^{-T} can only decrease it. This gives the same enclosure with C_b replaced by $C_{a,1}$. Summing the geometric Euler tail proves each enclosure. At noninteger indices its centers are not generally rational. The earlier sharp constant one applies to its stated range $a \geq 1$; the larger constants above are valid enclosures for the new range, without an optimality claim for the Euler remainder itself.

7 A proved local part of the normalized-radius conjecture

The earlier *Angular Zero Continuation* report [2] asks whether

$$\eta_{a,b}(\rho) = \frac{\cos \theta_{a,b}(\rho)}{\rho}$$

decreases on the whole interval $0 < \rho \leq 1$ for every integer $a \geq 2$, $b > 0$. At $a = 1$ it predicts decreasing behavior for $b > 1$, increasing behavior for $0 < b < 1$, and the constant $1/2$ for $b = 1$. That report identifies the sign of the first nonconstant Taylor coefficient as a smaller explicit open problem. We solve that problem for the full stated parameter range and describe the intervening fractional-order threshold.

Throughout this section put

$$U = 1 + 2^{-b}, \quad V = 1 + 2^{-b} + 3^{-b}, \quad W = 1 + 2^{-b} + 3^{-b} + 4^{-b}.$$

The parameter ρ in this section is a disk radius. It is unrelated to the Lerch deformation parameter used later.

Lemma 7.1 (The local radial coefficient). *For fixed $a \geq 1$, $b > 0$, $\eta_{a,b}$ extends as a real analytic even function near $\rho = 0$, and*

$$\eta_{a,b}(\rho) = \frac{U}{2} \left(\frac{2}{3}\right)^a + K(a,b)\rho^2 + O_{a,b}(\rho^4), \quad (7.1)$$

where

$$K(a,b) = UV 3^{-a} - \frac{W}{2} \left(\frac{2}{5}\right)^a - \frac{U^3}{2} \left(\frac{8}{27}\right)^a. \quad (7.2)$$

Proof. Set $\delta = \pi/2 - \theta$ and divide $\text{Im } F_{a,b}(\rho e^{i\theta})$ by $2^{-a}\rho^2$. The resulting equation is analytic near $(\rho, \delta) = (0, 0)$ and starts with $\sin(2\delta)$; its δ derivative there is 2. The implicit-function theorem gives

a unique analytic solution, odd under $(\rho, \delta) \mapsto (-\rho, -\delta)$. Writing $\delta = \alpha\rho + \beta\rho^3 + O(\rho^5)$, the modes with outer indices 2, 3, 4, 5 give

$$\alpha = \frac{U}{2}(2/3)^a, \quad \beta = UV 3^{-a} - \frac{W}{2}(2/5)^a - \frac{23U^3}{48}(8/27)^a.$$

All higher modes enter only at order ρ^5 or later in the normalized equation. Since $\eta = \sin \delta / \rho$, its quadratic coefficient is $\beta - \alpha^3/6$, which is (7.2). This also rederives the earlier report's local coefficient without assuming its sign. $\square$

Theorem 7.2 (Complete sign in the conjectured local range). *The coefficient (7.2) satisfies*

1. $K(a, b) < 0$ for every real $a \geq 2$ and every $b > 0$;
2. $K(1, b) < 0$ if $b > 1$, $K(1, b) > 0$ if $0 < b < 1$, and $K(1, 1) = 0$;
3. for $b \geq 1$, $K(a, b) < 0$ whenever $a > 1$;
4. for each $0 < b < 1$, there is exactly one $a_*(b) \in (1, 2)$ with $K(a_*(b), b) = 0$. The coefficient is positive on $1 \leq a < a_*(b)$ and negative for $a > a_*(b)$.

The function $a_*(b)$ is real analytic on $(0, 1)$. Every strict coefficient sign gives the corresponding strict monotonicity of $\eta_{a,b}$ on some interval $0 < \rho < \rho_0(a, b)$. No all-radius monotonicity claim is made here.

7.1 Monotonicity in the outer order

Define

$$G_a = W + U^3 \left(\frac{20}{27}\right)^a - 2UV \left(\frac{5}{6}\right)^a.$$

Then $K(a, b) = -\frac{1}{2}(2/5)^a G_a$, so it suffices to determine the sign of G_a . We first show

$$\frac{\partial G_a}{\partial a} > 0 \quad (a \geq 1, b > 0). \quad (7.3)$$

Put $x = 2^{-b}$ and $y = 3^{-b}$. Since $y \geq x^2$ and $0 < x < 1$,

$$\frac{U^2}{V} \leq \frac{(1+x)^2}{1+x+x^2} \leq \frac{4}{3}.$$

The last inequality is equivalent to $(1-x)^2 \geq 0$. Elementary alternating-series bounds for $\log(1+t)$ give

$$\log(27/20) < \frac{7273}{24000}, \quad \log(6/5) > \frac{9}{50}.$$

Consequently

$$\frac{U^2}{2V} \frac{\log(27/20)}{\log(6/5)} < \frac{7273}{6480} < \frac{9}{8} \leq \left(\frac{9}{8}\right)^a.$$

This is precisely the inequality obtained by differentiating G_a and moving its negative term to the other side. It proves (7.3).

7.2 A finite positive certificate at $a = 2$

Because $\log_2 3 > 3/2$, $y \leq x^{3/2}$. The coefficient of y in G_2 is $-(7 + 25x)/18 < 0$. With $x = t^2$, $0 < t < 1$, we obtain

$$G_2 \geq P_6(t) := \frac{800t^6 - 2025t^5 + 1833t^4 - 567t^3 - 192t^2 + 233}{1458}. \quad (7.4)$$

This polynomial has the degree-six Bernstein representation

$$P_6(t) = \sum_{j=0}^6 b_j \binom{6}{j} t^j (1-t)^{6-j},$$

with the exact coefficient vector

$$(b_0, \dots, b_6) = \left(\frac{233}{1458}, \frac{233}{1458}, \frac{367}{2430}, \frac{665}{5832}, \frac{55}{486}, \frac{95}{1458}, \frac{41}{729} \right). \quad (7.5)$$

Every coefficient is positive, and the Bernstein basis is nonnegative and sums to one on $[0, 1]$. Hence $G_2 > 0$. Together with (7.3), this proves the first assertion of the theorem. The accompanying verifier expands (7.5) over $\mathbb{Q}$ and checks equality with (7.4); no floating-point polynomial sign test is used.

7.3 The sign change at $a = 1$, $b = 1$

A direct simplification gives

$$27G_1 = 20x^3 + 42x^2 - 3x + 2 - (45x + 18)y. \quad (7.6)$$

If $b > 1$, let $t = \sqrt{2x} \in (0, 1)$. Since $y = (2x)^{\log_2 3}/3 \leq t^3/3$, substitution in (7.6) yields

$$G_1 \geq (1-t)P_5(t), \quad P_5(t) = \frac{-5t^5 + 10t^4 - 11t^3 + t^2 + 4t + 4}{54}.$$

The degree-five Bernstein coefficients of P_5 are

$$\left(\frac{2}{27}, \frac{4}{45}, \frac{19}{180}, \frac{14}{135}, \frac{1}{10}, \frac{1}{18} \right). \quad (7.7)$$

They are all positive, so $G_1 > 0$ for $b > 1$.

Suppose instead $0 < b < 1$, so $1/2 < x < 1$. Put $p = \log_2 3$. The exact integer inequality $2^{11} < 3^7$ gives $p > 11/7$, while $p < 2$. For $y(x) = x^p$ this implies

$$y(1/2) = \frac{1}{3}, \quad y'(1/2) > \frac{22}{21}, \quad y''(x) = p(p-1)x^{p-2} > \frac{44}{49} \quad (1/2 < x < 1).$$

Taylor's formula with an integral remainder therefore gives

$$y \geq \frac{1}{3} + \frac{22}{21}(x - 1/2) + \frac{22}{49}(x - 1/2)^2.$$

Using the negative coefficient of y in (7.6), we find

$$G_1 \leq -\frac{(2x-1)(10x^2 - 337x + 334)}{2646} < 0.$$

Indeed the quadratic is decreasing on $[1/2, 1]$ and its value at one is $7 > 0$. At $b = 1$, direct substitution $x = 1/2$, $y = 1/3$ gives $G_1 = 0$. The exact identity $F_{1,1}(z) = \frac{1}{2} \log^2(1 - z)$ gives $\eta_{1,1}(\rho) = 1/2$ for every $0 < \rho \leq 1$.

Finally, G_a is strictly increasing in $a \geq 1$. The signs of G_1 and G_2 give all the remaining assertions and the unique threshold. Its analyticity follows from the analytic implicit-function theorem and (7.3). Differentiating (7.1) shows that a nonzero K controls the sign of η' for sufficiently small positive ρ .

Remark 7.3 (What remains of the conjecture). This proves the local sign statement explicitly requested in the earlier report, including all real $a \geq 2$. It does not show that a derivative cannot change sign farther from the origin. At the fractional threshold $a = a_*(b)$, the quadratic coefficient vanishes; its quartic successor is needed even for local monotonicity. The next section proves opposite quartic signs at two inner orders, producing both maxima and minima. The full-radius integer conjecture and the general quartic-sign classification remain distinct unresolved questions.

8 A proved failure of fractional global monotonicity

The local sign threshold does not extend directly to a classification of radial monotonicity on the whole unit disk. There are fractional orders for which $\eta_{a,b}(\rho) = \cos \theta_{a,b}(\rho)/\rho$ first increases and then decreases. This is an analytic theorem, supported by a finite exact certificate for its hypotheses.

8.1 The next local coefficient

Put $p_n = H_{n-1}^{(b)}/n^a$ and

$$\mu = \frac{p_3}{2p_2}, \quad K = -\frac{4p_3\mu^2 - 4p_4\mu + p_5}{2p_2}.$$

Thus $\mu = \eta_{a,b}(0)$ and K is the previously identified quadratic coefficient. The next coefficient is

$$Q = -\frac{(8p_3\mu - 4p_4)K + 8p_4\mu^3 - 12p_5\mu^2 + 6p_6\mu - p_7}{2p_2}, \quad (8.1)$$

and the convergent local expansion is

$$\eta_{a,b}(\rho) = \mu + K\rho^2 + Q\rho^4 + O(\rho^6). \quad (8.2)$$

The coefficients and remainder are real analytic in a, b locally within the parameter range under consideration.

To check (8.1), write $c = \rho\eta$ and use $\sin(n\theta) = \sin \theta U_{n-1}(c)$. Division of the imaginary-part equation by $\rho^3 \sin \theta$ gives

$$\begin{aligned} 0 &= 2p_2\eta - p_3 + \rho^2(4p_3\eta^2 - 4p_4\eta + p_5) \\ &\quad + \rho^4(8p_4\eta^3 - 12p_5\eta^2 + 6p_6\eta - p_7) + O(\rho^6). \end{aligned}$$

Every omitted mode has order at least ρ^6 . This equation is analytic in $a, b, t = \rho^2, \eta$ near $t = 0$; it follows either from the convergent disk series or from the parity of each Chebyshev term. Its derivative in η at zero is $2p_2 > 0$. The analytic implicit-function theorem therefore gives (8.2), and substitution proves the formula for Q .

Theorem 8.1 (Emergence of a strict radial maximum). *Fix $b = 1/2$. There is a unique zero $a_\star$ of $K(a, 1/2)$ in the rational interval*

$$1.31298577512189 < a_\star < 1.31298577512190. \quad (8.3)$$

There exist $\delta > 0$ and $\rho_0 > 0$ such that, for every $a \in (a_\star - \delta, a_\star)$, the function $\eta_{a,1/2}(\rho)$ has exactly one critical point in $(0, \rho_0)$. It is a strict local maximum: the function increases before it and decreases after it within that interval. Writing this point as $\rho_{\max}(a)$, one has

$$\rho_{\max}(a) = \left(\frac{\partial_a K(a_\star, 1/2)}{2Q(a_\star, 1/2)} \right)^{1/2} (a_\star - a)^{1/2} + O((a_\star - a)^{3/2}). \quad (8.4)$$

The positive prefactor is rigorously enclosed by

$$44.0912731287 < \left(\frac{\partial_a K(a_\star, 1/2)}{2Q(a_\star, 1/2)} \right)^{1/2} < 44.0912738633.$$

In particular these fractional normalized-radius curves are not monotone on $(0, 1]$, despite having $K > 0$.

Proof. The accompanying standard-library verifier `certify_fractional_turning.py` uses only integer and rational arithmetic. Let I be the closed rational interval in (8.3). Its output proves

$$K(\inf I, 1/2) > 8.16 \cdot 10^{-17}, \quad K(\sup I, 1/2) < -8.72 \cdot 10^{-17},$$

and, throughout I ,

$$\begin{aligned} -0.016895831331 &< \partial_a K(a, 1/2) < -0.016895831329, \\ -4.345546 \cdot 10^{-6} &< Q(a, 1/2) < -4.345545 \cdot 10^{-6}. \end{aligned}$$

These signs alone give existence and uniqueness of $a_\star$ in I , independently of the more general local-threshold theorem. That theorem also identifies it as the unique threshold in $(1, 2)$.

Here is the complete arithmetic basis of the certificate. For $2 \leq n \leq 7$, it encloses $\log n$ by the first 120 terms of

$$\log n = 2 \sum_{j \geq 0} \frac{z^{2j+1}}{2j+1}, \quad z = \frac{n-1}{n+1},$$

with positive tail bounded by $2z^{241}/(241(1-z^2))$. For a nonnegative rational x , the first 60 terms after the constant term of e^x have tail at most

$$\frac{x^{61}}{61!} \frac{1}{1-x/62};$$

all arguments used here are less than three. Monotonicity encloses $e^{a \log n}$ at both ends of the interval arguments, and reciprocation encloses n^{-a} . The factors $j^{-1/2}$ in the harmonic numbers are bounded by exact integer-square-root operations. All arithmetic operations round outwards to multiples of 2^{-128} . Substitution in the formulas for K, Q and their elementary derivatives proves the displayed signs. Exact rational endpoints, including the tighter prefactor enclosure, are recorded in `fractional_turning_certificate.json`. The written tail bounds and integer inequalities make this a finite exact proof certificate; no floating-point sign decision is used.

Write $\eta_{a,1/2}(\rho) = \Phi(a, t)$, $t = \rho^2$, using the analytic function constructed above. Set $U(a, t) = \partial_t \Phi(a, t)$. Then

$$U(a_*, 0) = 0, \quad \partial_t U(a_*, 0) = 2Q(a_*, 1/2) < 0, \quad \partial_a U(a_*, 0) = \partial_a K(a_*, 1/2) < 0.$$

The analytic implicit-function theorem gives a unique local solution $t = t(a)$ of $U(a, t) = 0$, with

$$t(a) = \frac{\partial_a K(a_*, 1/2)}{2Q(a_*, 1/2)}(a_* - a) + O((a_* - a)^2).$$

It is positive for $a < a_*$ sufficiently close. On a sufficiently small fixed rectangle, $\partial_t U < 0$; at its upper t boundary, $U < 0$, while $U(a, 0) = K(a, 1/2) > 0$ for $a < a_*$. Thus there is exactly one such positive critical point and the derivative $\partial_\rho \eta = 2\rho U(a, \rho^2)$ changes from positive to negative there. At the critical point its second derivative is $4t\partial_t U < 0$. Taking the positive square root of the expansion for $t(a)$ proves (8.4). $\square$

8.2 Exploration beyond the existence theorem

A separate 70-digit diagnostic evaluates the defining disk series and its derivative at $b = 1/4, 1/2, 3/4$, at the local threshold and at offsets $-0.02, -10^{-4}, +0.02$, for radii 0.3, 0.7, 0.9, 0.98. For each b , the offset -10^{-4} gives a positive derivative at 0.3 and a negative derivative at 0.98. Thus the same phenomenon appears numerically at all three sampled inner orders; the theorem above proves it at $b = 1/2$.

For the exact rational pair $a = 21/16$, $b = 1/2$, the diagnostic finds

$$\eta'(0.9) = 1.2937181281564777 \cdot 10^{-6}, \quad \eta'(0.98) = -2.6349934635137804 \cdot 10^{-6}.$$

These two particular signs are numerical observations, not an extra certified instance of the theorem's unspecified interval in a . The disk-series truncation errors for F and zF' are bounded analytically by 10^{-50} , but rounding is not interval-enclosed. The existence theorem relies instead on the exact finite coefficient certificate above. Extending this argument to other b requires determining the sign of $Q(a_*(b), b)$. The complementary minimum theorem below proves that this sign is not always negative.

8.3 A complementary minimum and a change of quartic sign

The sign of the quartic coefficient along the local threshold also changes. In particular, the negative sign at $b = 1/2$ cannot extend to every $0 < b < 1$. The following exact certificate exhibits the opposite local bifurcation at a rational inner order close to one.

Theorem 8.2 (Emergence of a strict radial minimum). *Set $b_0 = 999/1000$. The unique local threshold $a_0 = a_*(b_0)$ satisfies the rational enclosure*

$$1.00057049936559087278 < a_0 < 1.00057049936559087279. \quad (8.5)$$

Moreover,

$$Q(a_0, b_0) > 0, \quad \partial_a K(a_0, b_0) < 0.$$

There exist $\delta > 0$ and $\rho_0 > 0$ such that, for every $a \in (a_0, a_0 + \delta)$, the function $\eta_{a,b_0}(\rho)$ has exactly one critical point in $(0, \rho_0)$. It is a strict local minimum: the function decreases before it and increases after it within this interval. Its location satisfies

$$\rho_{\min}(a) = \left(\frac{-\partial_a K(a_0, b_0)}{2Q(a_0, b_0)} \right)^{1/2} (a - a_0)^{1/2} + O((a - a_0)^{3/2}), \quad (8.6)$$

where the prefactor has the certified enclosure

$$17076.2396129683 < \left(\frac{-\partial_a K(a_0, b_0)}{2Q(a_0, b_0)} \right)^{1/2} < 17076.2396548521.$$

Thus fractional normalized-radius curves can also fail global monotonicity while their initial quadratic coefficient is negative. The theorem asserts a sufficiently small neighborhood; it gives no numerical lower bound on δ .

Proof. Let I be the closed rational interval in (8.5). The exact interval verifier `certify_fractional_minimum.py` proves

$$K(\inf I, b_0) > 4.08027 \cdot 10^{-23}, \quad K(\sup I, b_0) < -1.29944 \cdot 10^{-22}.$$

Throughout I , it also proves

$$\begin{aligned} 2.92779195 \cdot 10^{-11} &< Q(a, b_0) < 2.92779198 \cdot 10^{-11}, \\ -0.017074763285092 &< \partial_a K(a, b_0) < -0.017074763285091. \end{aligned}$$

The arithmetic uses the same outward dyadic intervals and rational logarithm/exponential remainder bounds as the maximum certificate. For this rational inner order, each j^{-b_0} is enclosed as the reciprocal of $e^{b_0 \log j}$, for $1 \leq j \leq 6$; each n^{-a} is enclosed in the same way, for $2 \leq n \leq 7$. All exponential arguments are less than three. Substitution into the finite formulas for K, Q and $\partial_a K$ establishes the signs. Exact rational endpoints and the sharper prefactor interval are in `fractional_minimum_certificate.json`. Thus both the small positive quartic coefficient and the much smaller endpoint values of K are certified without floating-point arithmetic.

Write $\eta_{a,b_0}(\rho) = \Phi(a, t)$ with $t = \rho^2$, and let $U = \partial_t \Phi$. At $(a_0, 0)$ one has

$$U = 0, \quad U_t = 2Q(a_0, b_0) > 0, \quad U_a = \partial_a K(a_0, b_0) < 0.$$

The analytic implicit-function theorem gives a unique local zero $t = t(a)$ with

$$t(a) = \frac{-\partial_a K(a_0, b_0)}{2Q(a_0, b_0)}(a - a_0) + O((a - a_0)^2).$$

It is positive for $a > a_0$ sufficiently close. On a sufficiently small fixed rectangle, $U_t > 0$; hence U has exactly one zero in the indicated positive t interval and changes from negative to positive there. Since $\eta_\rho = 2\rho U$, this is the asserted minimum, with $\eta_{\rho\rho} = 4tU_t > 0$ at the critical point. Taking its positive square root proves (8.6). $\square$

Corollary 8.3 (A zero of the quartic coefficient along the threshold). *There exists $b_\dagger \in (1/2, 999/1000)$ for which*

$$Q(a_\star(b_\dagger), b_\dagger) = 0.$$

Proof. The threshold equation has nonzero a derivative at every $a_\star(b)$, so $a_\star(b)$ is real analytic on $(0, 1)$. The finite formula for Q then makes $q(b) = Q(a_\star(b), b)$ real analytic there. The exact maximum and minimum certificates give $q(1/2) < 0 < q(999/1000)$, so the intermediate value theorem applies. $\square$

The finite-coefficient diagnostic `locate_quartic_sign_transition.py`, run at 100 decimal digits, locates one apparent sign change at

$$b_{\dagger} \approx 0.9974937898734204210746055931, \quad a_{\star}(b_{\dagger}) \approx 1.0014301809614951381293517294.$$

These displayed decimal locations are diagnostic, not certified enclosures. No uniqueness of $b_{\dagger}$ is claimed. The 11-point finite-coefficient sweep has negative $q(b)$ at every requested sample from 0.001 through 0.99, and positive $q(0.999)$; the latter sign is proved by the independent exact certificate. Determining the number of zeros of q and the higher-order radial behavior at its zeros remains a concrete research problem.

9 A Bessel scaling law when the Laurent index follows the derivative order

The fixed-index concentration theorem does not cover n comparable with k . In the adjacent-index regime $n = k + r$, with fixed integer r , the zeros close to $a = 1$ have a different scale and a different universal profile. Their distance from one is of order k^{-2} , the profile is a Bessel function, and the Lerch deformation changes these zeros only by an exponentially small amount. The result holds uniformly up to the classical endpoint $\rho = 1$.

Throughout, let

$$F_{n,k}^{\rho}(a) = (-1)^{n+k} \mathcal{D}_{n,k}^{\rho}(a), \quad f_{n,k}(a) = F_{n,k}^0(a).$$

The exact coefficient and shift identities are

$$f_{n,k}(a) = \frac{n!}{a^{k+1}} [t^n] (1+t)_k e^{-t \log a}, \quad F_{n,k}^{\rho}(a) = \sum_{m \geq 0} \rho^m f_{n,k}(a+m). \quad (9.1)$$

The shift series converges absolutely, also at $\rho = 1$, for $k \geq 1$. Here $(1+t)_k = \prod_{j=1}^k (t+j)$, so the coefficient polynomial in $\log a$ has integer coefficients. It is not a reciprocal-gamma Appell polynomial.

9.1 An exponentially small spectral tail

The following estimate retains the angle in Cauchy's coefficient bound. Replacing $|j+t|$ by $j+|t|$ before optimizing loses the useful decay.

Lemma 9.1 (A contour estimate in the adjacent-index range). *For integers $n \geq 0$, $k \geq 1$, $a \geq 3/4$ and $0 \leq \rho \leq 1$,*

$$\frac{|F_{n,k}^{\rho}(a) - f_{n,k}(a)|}{n!} \leq e^7 \left(\frac{k}{2}\right)^{k-n} e^{-k/20} \left(\frac{4}{7} + \frac{2}{k}\right). \quad (9.2)$$

More generally, for fixed nonnegative integers p, q , derivatives $\partial_{\rho}^p \partial_a^q$ of the spectral tail satisfy

$$\frac{|\partial_{\rho}^p \partial_a^q (F_{n,k}^{\rho} - f_{n,k})(a)|}{n!} \leq C_p \left(\frac{k+q}{2}\right)^{k+q-n} e^{-(k+q)/20}$$

for all sufficiently large k (depending on p), uniformly for $a \geq 3/4$ and $\rho \in [0, 1]$. At $\rho = 1$ these are one-sided derivatives when needed.

Proof. Put $R = k/2$, $t = Re^{i\theta}$ and $v = \cos \theta$. By the arithmetic–geometric mean inequality,

$$\begin{aligned} \frac{|(1+t)_k|}{R^k} &\leq \left(\frac{1}{kR^2} \sum_{j=1}^k (j^2 + R^2 + 2jRv) \right)^{k/2} \\ &= (A_k + B_kv)^{k/2}, \\ A_k &= \frac{7}{3} + \frac{2}{k} + \frac{2}{3k^2}, \quad B_k = 2 + \frac{2}{k}. \end{aligned}$$

For $b \geq b_0 = 7/4$ and $L = \log b$, Cauchy's bound in (9.1) therefore gives

$$\frac{|f_{n,k}(b)|}{n!} \leq R^{k-n} b^{-1} \exp \left\{ k \max_{-1 \leq v \leq 1} \left(\frac{1}{2} \log(A_k + B_kv) - L(1 + v/2) \right) \right\}. \quad (9.3)$$

Write $L_0 = \log(7/4)$. Since $1 + v/2 \geq 1/2$, replacing L by L_0 costs at most $-(L - L_0)/2$ in the displayed exponent. Moreover,

$$\frac{1}{2} \log(A_k + B_kv) \leq \frac{1}{2} \log(7/3 + 2v) + \frac{7}{k}.$$

Indeed $7/3 + 2v \geq 1/3$, while the nonnegative increment is at most $4/k + 2/(3k^2)$; apply $\log(1+x) \leq x$. If $y = 7/3 + 2v$, maximizing over all $y > 0$ gives the upper bound

$$\frac{1}{2} \log \frac{2}{L_0} - \frac{1}{2} - \frac{5}{12} L_0 < -\frac{1}{20}.$$

For an entirely elementary verification, use $L_0 > 11/20$ and $\log(40/11) < 13/10$. The expression is then smaller than $13/20 - 1/2 - 11/48 = -19/240 < -1/20$. These logarithm bounds follow, for example, by the positive atanh series for the logarithm with a geometric remainder. Consequently

$$\frac{|f_{n,k}(b)|}{n!} \leq e^7 R^{k-n} e^{-k/20} b^{-1} (b_0/b)^{k/2}.$$

The function $b^{-1}(b_0/b)^{k/2}$ decreases. Summing it over $b = a + m$, $m \geq 1$, and comparing the tail with its integral gives at most $b_0^{-1} + 2/k = 4/7 + 2/k$, which proves (9.2).

For derivatives, $\partial_a^q f_{n,k} = (-1)^q f_{n,k+q}$, and $\partial_\rho^p \rho^m = m^p \rho^{m-p}$. The latter factor is bounded by $m^p \leq (a+m)^p$ on the relevant range. For $K = k + q > 2p$, the resulting scalar sum is bounded by

$$b_0^{p-1} + \frac{b_0^p}{K/2 - p}.$$

The same estimate proves uniform convergence of the differentiated series and hence justifies the derivatives, including at $\rho = 1$. $\square$

9.2 The limiting entire profile and its complete expansion

Fix an integer r , and take k large enough that $n = k + r \geq 0$. Set

$$a = e^{z/k^2}, \quad \mathcal{H}_{r,k}^\rho(z) = \frac{k^{2r} a^{k+1}}{(k+r)!} F_{k+r,k}^\rho(a). \quad (9.4)$$

The entire function defining the limit is

$$\Psi_r(z) = \sum_{j \geq \max(0, -r)} \frac{(-z)^{r+j}}{2^j j! (r+j)!}. \quad (9.5)$$

For $z > 0$, the standard series for the Bessel function gives

$$\Psi_r(z) = (-1)^r (2z)^{r/2} J_r(\sqrt{2z}). \quad (9.6)$$

The series (9.5), rather than the separate powers in (9.6), defines the value at zero and the entire continuation. Since $J_{-s} = (-1)^s J_s$ for integer s , its positive zeros are $j_{|r|,m}^2/2$, $m \geq 1$. They are simple. The standard Bessel series and its zero properties are recorded in DLMF §§10.2 and 10.21 [8]; these classical facts are not new claims here.

Theorem 9.2 (Uniform Bessel profile). *For fixed $r \in \mathbb{Z}$, on every compact real z interval,*

$$\mathcal{H}_{r,k}^\rho(z) = \Psi_r(z) + \frac{1}{3k} ((2r+5)z\Psi_r'(z) + (z - r(2r+5))\Psi_r(z)) + O(k^{-2}), \quad (9.7)$$

uniformly for $\rho \in [0, 1]$, including every fixed number of z derivatives. There is a complete expansion in integral powers of k^{-1} . Every coefficient in that expansion is independent of ρ . The difference between the profile at any ρ and at zero is $O(k^r e^{-k/20})$, with the same bound after any fixed number of ρ derivatives and a fixed number of z derivatives.

Proof. Let $E_{j,k} = e_j(1, \dots, k)$ be the elementary symmetric functions of the integers $1, \dots, k$. Reversing the coefficients of the rising factorial gives

$$(1+t)_k = \sum_{j=0}^k E_{j,k} t^{k-j}.$$

Consequently the elementary profile is exactly

$$\mathcal{H}_{r,k}^0(z) = \sum_{j=\max(0,-r)}^k \frac{E_{j,k}}{k^{2j}} \frac{(-z)^{r+j}}{(r+j)!}. \quad (9.8)$$

Consider the auxiliary polynomial

$$E_k(t) = \sum_{j=0}^k \frac{E_{j,k}}{k^{2j}} t^j = \prod_{\ell=1}^k \left(1 + \frac{\ell t}{k^2}\right).$$

Its logarithm has the convergent local expansion

$$\begin{aligned} \log E_k(t) &= \sum_{m \geq 1} \frac{(-1)^{m-1} t^m}{m k^{2m}} \sum_{\ell=1}^k \ell^m \\ &= \frac{t}{2} + \frac{t/2 - t^2/6}{k} + \frac{-t^2/4 + t^3/12}{k^2} + O(k^{-3}). \end{aligned} \quad (9.9)$$

The remainder is uniform on each fixed complex t disk. For any desired order, the terms with large m have a uniform geometric bound because $|\ell t/k^2| \leq |t|/k$. Each retained power sum is an exact polynomial in k , giving a complete expansion. Exponentiating,

$$E_k(t) = e^{t/2} \left(1 + \frac{t/2 - t^2/6}{k} + \frac{-t^2/8 + t^4/72}{k^2} + O(k^{-3})\right). \quad (9.10)$$

Cauchy's coefficient estimate on a fixed circle bounds each coefficient remainder by $C k^{-N-1} T^{-j}$. Multiplying by $(-z)^{r+j}/(r+j)!$ and summing proves a locally uniform expansion of (9.8), with all

derivatives. This step controls the infinite coefficient sum; coefficientwise convergence alone would not suffice.

Put $D = z\partial_z - r$. The coefficient of k^{-1} is

$$\frac{5D - 2D^2}{3} \Psi_r.$$

The series (9.5) satisfies

$$z\Psi_r'' + (1 - r)\Psi_r' + \frac{1}{2}\Psi_r = 0. \quad (9.11)$$

Using this equation reduces the last differential expression to the coefficient displayed in (9.7). The second coefficient, useful below, is

$$\left(\frac{2}{9}D(D-1)(D-2)(D-3) - \frac{1}{2}D(D-1) \right) \Psi_r. \quad (9.12)$$

The remaining coefficients follow by the same finite process from (9.9).

Finally, on a compact z interval, $a = e^{z/k^2} \geq 3/4$ for large k , and a^{k+1} is uniformly bounded. The spectral-tail lemma, multiplied by k^{2r} , gives the error $O(k^r e^{-k/20})$. Its differentiated bounds give the same result for fixed mixed derivatives: each z derivative introduces factors of k^{-2} when differentiating a , while an additional a derivative of the tail costs only a factor $O(k)$. Derivatives of a^{k+1} cost $O(k^{-1})$ in the z coordinate. Thus all these errors remain exponentially small and affect none of the algebraic coefficients. $\square$

9.3 Simple zeros and deformation independence to every algebraic order

Theorem 9.3 (Bessel locations of adjacent-index zeros). *Fix $r \in \mathbb{Z}$ and $m \geq 1$, and put $\lambda = j_{|r|,m}^2/2$. For all sufficiently large k , every $\rho \in [0, 1]$ has exactly one zero $a_{r,m,k}^\rho$ in a fixed sufficiently small neighborhood of $k^2 \log a = \lambda$. The zero is positive and simple. Uniformly in ρ ,*

$$k^2 \log a_{r,m,k}^\rho = \lambda - \frac{(2r+5)\lambda}{3k} + \frac{\lambda(4r^2 + 15r + \lambda + 20)}{9k^2} + O(k^{-3}), \quad (9.13)$$

$$a_{r,m,k}^\rho = 1 + \frac{\lambda}{k^2} - \frac{(2r+5)\lambda}{3k^3} + \frac{\lambda(8r^2 + 30r + 11\lambda + 40)}{18k^4} + O(k^{-5}). \quad (9.14)$$

Both formulas extend to all orders by a recursively computable expansion with coefficients independent of ρ . In addition,

$$\sup_{0 \leq \rho \leq 1} |a_{r,m,k}^\rho - a_{r,m,k}^0| = O(k^{r-2} e^{-k/20}). \quad (9.15)$$

For every fixed positive integer p , the same bound holds for $|\partial_\rho^p a_{r,m,k}^\rho|$, once k is sufficiently large. No assertion about the total positive-zero count follows from this local scaling theorem.

Proof. The positive Bessel zero λ is simple. Choose a fixed small closed interval about it on which Ψ_r' never vanishes and whose endpoints have opposite signs. Uniform C^1 convergence in Theorem 9.2 gives exactly one simple real zero there for all sufficiently large k , uniformly in ρ .

Substitute $z = \lambda + d/k + e/k^2$ into the profile expansion. The k^{-1} coefficient gives $d = -(2r+5)\lambda/3$. At a zero of Ψ_r , equation (9.11) gives

$$\frac{\Psi_r''(\lambda)}{\Psi_r'(\lambda)} = \frac{r-1}{\lambda}.$$

Writing H_1 for the k^{-1} coefficient in the profile, direct differentiation and the same differential equation give $H'_1(\lambda)/\Psi'_r(\lambda) = \lambda/3$. Formula (9.12) gives

$$\frac{H_2(\lambda)}{\Psi'_r(\lambda)} = -\frac{2r^3\lambda}{9} - \frac{4r^2\lambda}{3} + \frac{2r\lambda^2}{9} - \frac{35r\lambda}{18} + \frac{4\lambda^2}{9} - \frac{5\lambda}{6}.$$

Therefore

$$e = -\frac{\Psi''_r(\lambda)d^2/2 + H'_1(\lambda)d + H_2(\lambda)}{\Psi'_r(\lambda)} = \frac{\lambda(4r^2 + 15r + \lambda + 20)}{9}.$$

This proves (9.13); exponentiation gives (9.14). At every further order the next unknown coefficient has the same nonzero multiplier $\Psi'_r(\lambda)$. The complete profile expansion and a uniform lower bound for the derivative turn this formal recursion into an asymptotic expansion with the stated remainders, by the mean value theorem.

The profile difference between ρ and zero is $O(k^r e^{-k/20})$. Dividing by the uniform derivative lower bound gives the same error in z . The map $a = e^{z/k^2}$ supplies the additional factor k^{-2} in (9.15). For derivatives in ρ , use the mixed-derivative bounds of the tail lemma and differentiate the implicit equation. Every positive ρ derivative of the profile is exponentially small; its z derivative stays bounded away from zero. Induction proves the same bound for every fixed root derivative. The required derivatives at $\rho = 1$ are justified by the uniformly convergent differentiated shift series, for sufficiently large k . $\square$

For $r = 0$, the limiting function is $J_0(\sqrt{2z})$. The first zero thus satisfies, for every Lerch parameter including one,

$$a = 1 + \frac{j_{0,1}^2}{2k^2} - \frac{5j_{0,1}^2}{6k^3} + O(k^{-4}).$$

In particular this locates a simple zero of $\gamma_k^{(k)}$ near one. More generally, the classical specialization of the profile theorem can be written without the sign-normalized family:

$$\frac{k^{2r} e^{(k+1)z/k^2}}{(k+r)!} \gamma_{k+r}^{(k)}(e^{z/k^2}) \longrightarrow (2z)^{r/2} J_r(\sqrt{2z}), \quad z > 0.$$

For $r > 0$ the limiting profile also has a zero of multiplicity r at $z = 0$. The theorem deliberately treats the positive Bessel zeros, not this multiple central zero. Its finer unfolding is sensitive to the exponentially small spectral tail and is not determined by the algebraic expansion.

10 The full small-Lerch-parameter unfolding at arbitrary derivative order

The earlier report [3] proves the small- ρ count for every n at $k = 1$. The following generalization handles every pair (n, k) . It also explains why a parity-only count does not extend to arbitrary derivative order: an explicit rational polynomial evaluated at $\log 2$ controls the answer.

Theorem 10.1 (All-pair small- ρ count). *Fix integers $n \geq 1$ and $k \geq 1$. If $n \leq k + 1$, then $F_{n,k}^\rho$ has exactly n simple positive zeros for all sufficiently small $\rho \geq 0$.*

If $n > k$, put $r = n - k$ and $B_{n,k} = f_{n,k}(2)$. Then $B_{n,k} \neq 0$, and, for all sufficiently small $\rho > 0$, the number of positive zeros is exactly

$$\begin{cases} k+1, & r \text{ odd}, \\ k+2, & r \text{ even and } B_{n,k} < 0, \\ k, & r \text{ even and } B_{n,k} > 0. \end{cases} \quad (10.1)$$

All these zeros are simple. The k noncentral zeros continue analytically from the elementary endpoint. Every real central branch has

$$a(\rho) = 1 + u\rho^{1/r} + O(\rho^{2/r}), \quad u^r = -\frac{(-1)^r r!}{n!} B_{n,k}. \quad (10.2)$$

The real solutions of the displayed equation give exactly the real central branches. When $r = 1$, the first and second assertions agree.

Proof. At $\rho = 0$, the polynomial in $x = -\log a$ is a positive scalar multiple of $\prod_{j=1}^k (1 + \partial_x/j)x^n$. Each application of $1 + c\partial_x$, $c > 0$, preserves real roots, decreases every inherited multiple-root multiplicity by one, and creates simple roots strictly between consecutive distinct roots and to the left of the least root. Hence, for $n > k$, there are k distinct negative simple roots and a root at zero of multiplicity exactly $r = n - k$. For $n \leq k + 1$ every root is simple. The corresponding noncentral positive a roots persist analytically for small ρ .

The polynomial

$$n![t^n](1+t)_k e^{-Lt} \in \mathbb{Z}[L]$$

is nonzero (its leading term is $k!(-L)^n$). By Lindemann's classical theorem [9], $\log 2$ is transcendental: if it were nonzero algebraic, its exponential could not equal the algebraic number two. It cannot be a zero of this nonzero integer polynomial. Thus $B_{n,k} \neq 0$. This arithmetic argument applies to the elementary coefficient polynomial; it would not apply to a polynomial with uncontrolled gamma or zeta constants as coefficients.

Writing $a = 1 + \delta$ and using $\log^n(1 + \delta)/(1 + \delta) = \delta^n + O(\delta^{n+1})$ gives

$$F_{n,k}^\rho(1 + \delta) = \frac{(-1)^r n!}{r!} \delta^r + \rho B_{n,k} + O(\delta^{r+1} + \rho|\delta| + \rho^2).$$

Set $\rho = \varepsilon^r$ and $\delta = \varepsilon u$. After division by ε^r , the limiting equation is (10.2). Its r complex roots are distinct and nonzero. The analytic implicit function theorem gives r branches analytic in ε ; Rouché's theorem on a small fixed δ disk shows that these exhaust the roots tending to one. For real ε , branches with real initial values remain real and the others remain nonreal. For odd r there is one real root. For even r , there are two real roots precisely when the right-hand side is positive, that is, when $B_{n,k} < 0$; otherwise there are none. Each real branch is simple for small positive ε .

To exclude further positive roots, take A larger than the largest elementary root. All $f_{n,k}(a + m)$ have the common sign $(-1)^n$ for $a > A$, so there are no roots there. For sufficiently small a , the positive divergent elementary term dominates the remaining series, uniformly for $0 \leq \rho \leq 1/2$. On the compact set left after removing the elementary root neighborhoods, uniform convergence to the nonzero elementary function excludes zeros. This proves both global completeness and the count. $\square$

Corollary 10.2 (A necessary condition for uniform positive-parameter saturation). *For $1 \leq k \leq n - 3$, saturation by n positive zeros cannot hold for every $\rho \in (0, 1]$. At the borderline $k = n - 2$, saturation holds for all sufficiently small positive ρ if and only if $f_{n,n-2}(2) < 0$.*

Proof. For $r \geq 3$, the count in (10.1) is strictly smaller than $k + r = n$. For $r = 2$, it is n precisely for the negative sign, and otherwise equals $n - 2$. $\square$

The following are finite exact-arithmetic consequences, separate from the all-pair analytic theorem. Each sign is certified by a rational interval for $f_{n,k}(2)/n!$ in the supplied

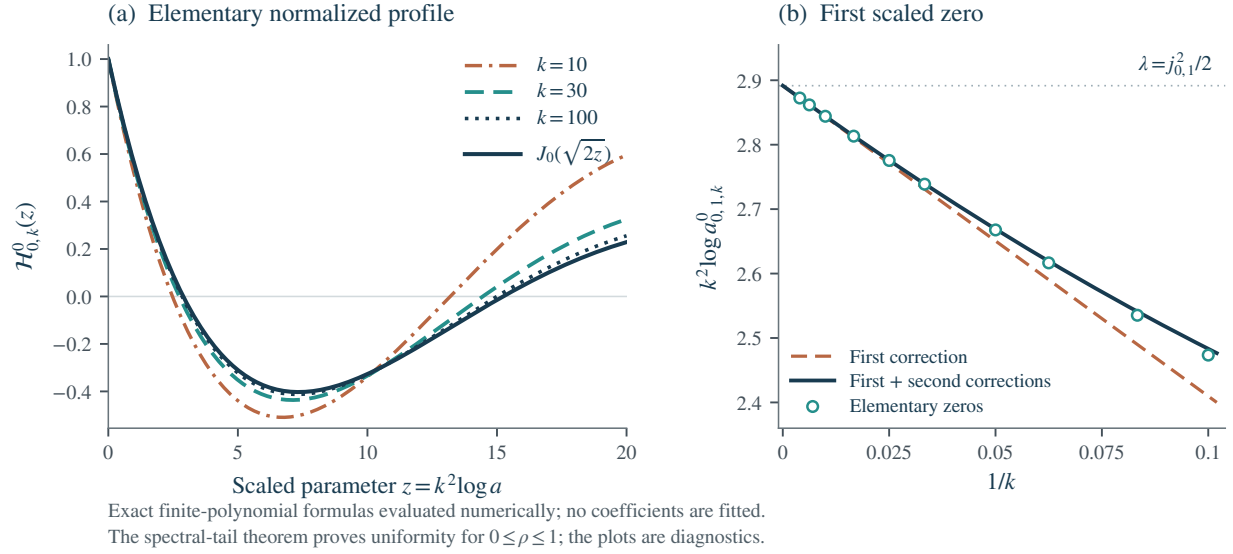


Figure 2: The adjacent-index Bessel regime at $r = 0$. Left: exact elementary profile polynomials evaluated numerically and their limiting Bessel function. Right: the first elementary scaled zero and the proved first two corrections. No coefficients are fitted. The uniform spectral-tail theorem, rather than the plots, establishes the result for $0 \leq \rho \leq 1$.

`evidence/exact-signs.json`; the code uses the integer rising factorial and a rational logarithm enclosure.

n	k	Sign of $f_{n,k}(2)$	Count for sufficiently small $\rho > 0$
4	2	+	2
5	3	+	3
6	2	+	2
6	4	−	6
7	5	−	7
8	6	+	6

The interval of sufficiently small positive parameters depends on (n, k) and is not numerically specified by this table. These counts make no assertion about an intermediate Lerch parameter or the classical endpoint. The separate full-Lerch Bessel diagnostics in the package are floating-point observations and are not used in any sign certificate or global count.

Further problems made precise. The central multiple zero in the Bessel scaling theorem lies on a smaller, exponentially sensitive scale. Its joint k, ρ unfolding, including possible cancellations between spectral terms, is a natural next problem. A second problem is a uniform expansion when $|n - k|$ grows, which should connect the fixed-order Bessel regime to a large-order Bessel or Airy regime. Neither follows from the present fixed- r theorem. A third problem is the oscillatory sign pattern of $f_{n,k}(2)$ when n and k grow together; the all-pair small- ρ classification reduces that question to an explicit finite polynomial, but does not establish an asymptotic sign law.

11 A transition for two growing log-gamma exponents

Write $f(x) = \log \Gamma(x)$, which is positive for $0 < x < 1$, and extend the reflected moments to positive real exponents:

$$M_{n,m} = \int_0^1 f(x)^n f(1-x)^m dx.$$

The moments themselves, their reflection symmetry, and associated exact identities are classical [10]. The pinned manuscript proves, for every *fixed* nonnegative integer m ,

$$M_{n,m} \sim \frac{\gamma^m \Gamma(n+1)}{(m+1)^{n+1}} \quad (n \rightarrow \infty), \quad (11.1)$$

and develops a considerably sharper fixed- m theory. It explicitly leaves simultaneous growth as a separate problem. Here we locate and resolve a transition in which (11.1) no longer has relative error tending to zero.

Set

$$A = \frac{\zeta(2)}{2\gamma}, \quad B = \frac{\zeta(3)}{3\gamma} - \frac{\zeta(2)^2}{8\gamma^2}, \quad C = A - \gamma. \quad (11.2)$$

Thus $C = 0.84767122475766887678956780929 \dots > 0$. Only the exact definitions enter the proofs. Positivity also follows without decimals: $\gamma < H_5 - \log 5 < 7/10$, using the first three positive terms of $2 \operatorname{arctanh}(2/3)$ to bound $\log 5$ from below, while $\zeta(2) > 1 > 2(7/10)^2$.

Theorem 11.1 (Critical reflected-moment transition). *Let c range over a fixed compact subset of $\mathbb{R}$. As m tends to infinity through positive real values, put*

$$L = \log m + c, \quad q = e^{-c}, \quad n = (m+1)L.$$

Then, uniformly in c ,

$$\frac{(m+1)^{n+1} M_{n,m}}{\gamma^m \Gamma(n+1)} = e^{Cq} \left\{ 1 + \frac{P_1(L, q)}{m} + O\left(\frac{L^2}{m^2}\right) \right\}, \quad (11.3)$$

where

$$P_1(L, q) = \frac{L}{2}(Cq + C^2 q^2) - (A + \gamma)q + (B + \gamma^2)q^2. \quad (11.4)$$

In particular, if integer pairs (n, m) satisfy $n/(m+1) - \log m \rightarrow c$, their normalized moments converge to $\exp(Ce^{-c})$.

For example, at $c = 0$ the limiting ratio is $e^C = 2.3342046795191067941 \dots$. This is an explicit counterexample to an unrestricted extension of (11.1) to a growing reflected exponent. It is not a counterexample to the manuscript, which keeps m fixed in that assertion.

11.1 The exact gamma expectation

The local series needed below follow from the usual expansion of $\log \Gamma(1+z)$ [7]. Define

$$p(x) = \log \Gamma(1+x), \quad h(x) = \log \left(\frac{\log \Gamma(1-x)}{\gamma x} \right).$$

Both are real analytic near zero, with

$$\begin{aligned} p(x) &= -\gamma x + \frac{\zeta(2)}{2}x^2 + O(x^3), \\ h(x) &= Ax + Bx^2 + O(x^3). \end{aligned} \quad (11.5)$$

Substitution $x = e^{-t}$ gives an exact identity. If T has gamma density

$$\frac{(m+1)^{n+1}}{\Gamma(n+1)} t^n e^{-(m+1)t}, \quad t > 0, \quad (11.6)$$

then the left side of (11.3) is

$$\mathbb{E} e^{H_m(T)}, \quad H_m(t) = n \log \left(1 + \frac{p(e^{-t})}{t} \right) + m h(e^{-t}). \quad (11.7)$$

The logarithm is real: $t + p(e^{-t}) = f(e^{-t}) > 0$. The base approximation treats $f(e^{-t})$ as t and $f(1 - e^{-t})$ as γe^{-t} . Their first corrections act together at the scale $m e^{-L} = q$.

Lemma 11.2 (Localization with the full weight). *Fix a compact set of c values and $\delta \in (0, 1/2)$. Under the hypotheses of Theorem 11.1, for every fixed $N > 0$,*

$$\mathbb{E} \left[e^{H_m(T)} \mathbf{1}_{\{|T-L|>\delta\}} \right] = O_N(m^{-N}).$$

The same superpolynomial bound holds for the unweighted gamma expectations of any fixed power of $|T - L|$ on this complement.

Proof. All constants below may depend on the compact set of c values and on δ . First, $p(x) < 0$ for $0 < x < 1$: strict log-convexity of Γ between 1 and 2 gives $\Gamma(1+x) < 1$. Thus

$$e^{H_m(t)} \leq e^{m h(e^{-t})}.$$

On $t \geq L/2$, the Taylor expansion of h bounds this by $\exp(K m e^{-t}) \leq \exp(K' \sqrt{m})$. The gamma moment-generating function is $(1 - v/(m+1))^{-(n+1)}$. Its Chernoff bounds give

$$\Pr(|T - L| > \delta) \leq 2 \exp(-c_\delta m/L)$$

for large m . Since $m/L \gg \sqrt{m}$, the contribution of this part of the complement is smaller than every inverse power of m .

Choose a fixed $\varepsilon \in (0, 1/2)$. On $\varepsilon \leq t \leq L/2$, $h(e^{-t})$ is bounded above by a constant. The same gamma Chernoff calculation at $T \leq L/2$ gives $\Pr(T \leq L/2) \leq \exp(-c_0 m L)$, with $c_0 > 0$; multiplying by $e^{K m}$ preserves superpolynomial decay.

For $0 < t < \varepsilon$, use the elementary endpoint bounds

$$f(e^{-t}) \leq K t, \quad f(1 - e^{-t}) \leq D + |\log t|,$$

with fixed positive K, D . The contribution before normalization is at most

$$K^n \int_0^\varepsilon t^n (D - \log t)^m dt \leq \frac{K^n D^m}{n+1 - m/D}.$$

Here $(D + y)^m \leq D^m e^{m y/D}$ was used, and the denominator is positive for large m . Stirling's estimate shows that division by $\gamma^m \Gamma(n+1)/(m+1)^{n+1}$ makes the logarithm of this bound at most $-n \log L + O(n+m)$. This again is smaller than every inverse power. The unweighted polynomial-tail assertions follow from the same gamma bounds, or from their moment-generating function after a fixed small exponential tilt. $\square$

11.2 Proof of the transition and its first correction

Proof of Theorem 11.1. Put $U = T - L$. Its first raw moments about L , directly from (11.6), are

$$\begin{aligned}\mathbb{E}U &= \frac{1}{m+1}, & \mathbb{E}U^2 &= \frac{L}{m+1} + \frac{2}{(m+1)^2}, \\ \mathbb{E}U^3 &= \frac{5L}{(m+1)^2} + \frac{6}{(m+1)^3}, & \mathbb{E}U^4 &= O(L^2/m^2).\end{aligned}\tag{11.8}$$

The coefficient 5 in the third raw moment includes the shift between L and the mean $L + 1/(m+1)$.

On $|u| \leq \delta$, the functions $e^{H_m(L+u)}$ and their first four derivatives are uniformly bounded. Indeed $e^{-L} = q/m$, $L/(L+u)$ is bounded, and the germs in (11.5) can be differentiated uniformly there. To leading order write

$$H_0(u) = qe^{-u} \left(A - \frac{\gamma L}{L+u} \right).$$

The values required through order m^{-1} are

$$\begin{aligned}H_m(L) &= Cq + \frac{1}{m} \left\{ -\gamma q + q^2 \left(B + \frac{\zeta(2)}{2} - \frac{\gamma^2}{2L} \right) \right\} + O(m^{-2}), \\ H'_0(0) &= q(-C + \gamma/L), \\ H''_0(0) &= q(C - 2\gamma/L - 2\gamma/L^2).\end{aligned}\tag{11.9}$$

The first two derivatives of $H_m(L+u)$ differ from the corresponding derivatives of $H_0(u)$ by $O(m^{-1})$.

Taylor-expand $e^{H_m(L+u)}$ through the cubic term, with a uniform $O(u^4)$ remainder. Lemma 11.2 permits use of the full moments (11.8). The cubic contribution is $O(L/m^2)$ and the quartic remainder is $O(L^2/m^2)$. Therefore

$$\begin{aligned}e^{-Cq} \mathbb{E}e^{H_m(T)} &= 1 + \frac{1}{m} \left[-\gamma q + q^2 \left(B + \frac{\zeta(2)}{2} - \frac{\gamma^2}{2L} \right) + H'_0(0) \right. \\ &\quad \left. + \frac{L}{2} \{ H''_0(0) + H'_0(0)^2 \} \right] + O(L^2/m^2).\end{aligned}$$

Inserting (11.9) cancels every displayed negative power of L . Since $\gamma A = \zeta(2)/2$, the remaining coefficient is exactly (11.4). This proves the uniform expansion. The assertion for integer pairs follows by using their actual $c_m = n/(m+1) - \log m$ in the uniform formula. $\square$

11.3 Higher orders and the early-pole resummation

The proof is effective at every finite order. One expands the two analytic germs to the required degree, expands in u on the localized interval, and uses exact gamma moments. For any fixed R , taking the Taylor expansion in u through degree $2R+1$ bounds its next remainder by $O((L/m)^{R+1})$. The terms from odd moments must be kept with their signs; replacing all of them by absolute-moment bounds would lose powers of m .

There is a useful algebraic description of the first correction. Let $p_1 = -\gamma$, $p_2 = \zeta(2)/2$, and $s_2 = p_2 + B$. The formal early-pole coefficients are

$$C_j(m) = [x^j] \exp\{(m+j+1)p(x) + mh(x)\}.$$

For fixed j , $C_j(m) = m^j C^j / j! + O_j(m^{j-1})$, whereas

$$\left(\frac{m+1}{m+j+1} \right)^{n+1} = \left(\frac{q}{m} \right)^j \left\{ 1 + \frac{Lj^2/2 - j}{m} + O_j(L^2/m^2) \right\}.$$

Thus the leading formal resummation is $\sum_{j \geq 0} (Cq)^j / j! = e^{Cq}$. With $\mathcal{D} = q\partial_q$, the next coefficient is

$$e^{Cq} P_1(L, q) = \left\{ s_2 q^2 + p_1 q (\mathcal{D} + 2) + \frac{L}{2} \mathcal{D}^2 - \mathcal{D} \right\} e^{Cq}. \quad (11.10)$$

This identity is checked by ordinary differentiation and provides an independent symbolic derivation of (11.4). It is not an instruction to sum the manuscript's divergent infinite pole series. The localization proof above controls the actual integral and supplies the analytic justification for the displayed transition.

11.4 Location of the transition on the integer lattice

The scale is $m \asymp n / \log n$. More precisely, if c is fixed and $w = W(e^c(n+1))$, put $k_0 = (n+1)/w$. The positive real solution of $n = (m+1)(\log m + c)$ satisfies

$$m = k_0 - 1 + \frac{1}{2k_0(w+1)} + O\left(\frac{1}{k_0^2(w+1)}\right). \quad (11.11)$$

To check this, write $k = m+1$ and expand

$$n = k(\log k + c) - 1 - \frac{1}{2k} + O(k^{-2}).$$

The equation $k_0(\log k_0 + c) = n+1$ and one Taylor expansion, whose derivative is $w+1$, give (11.11). Nearest-integer choices may then be inserted using their actual value of c_m .

The transition settles a concrete part of the simultaneous-exponent problem. It does not classify the regime $m/n \rightarrow \lambda > 0$, nor the uniform least-term remainder when both m and the truncation index grow. Those require additional saddle and inverse-function estimates.

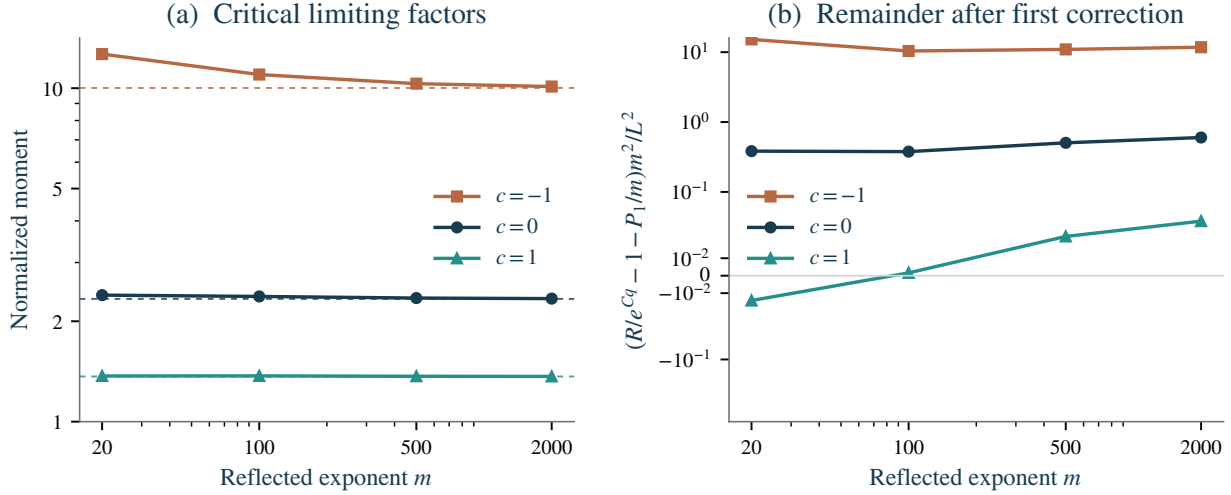


Figure 3: Numerical diagnostics for the joint gamma-moment transition. Left: normalized moments tend to different nonunit limits as m grows. Right: the relative error after the first correction, multiplied by m^2/L^2 , is consistent with the proved uniform remainder order. The displayed values are high-precision quadrature, not interval certificates. Six values were independently integrated in both x and $t = -\log x$.

12 The exact reach of convergent Cayley symmetry

The existing S_4 proof uses the involution $\phi(t) = (1-t)/(1+t)$ together with double shuffle. Before attempting a larger combined relation matrix, one can answer an exact structural question: how much does this convergent involution prove by itself, and how much stronger does it become after all shuffle products of its relations are admitted? These are two different presentations. Both can be computed in every weight.

The resulting dimensions below are dimensions of explicitly defined formal word quotients. They are not dimensions of evaluated periods. The theorem includes neither stuffle nor distribution relations, so it does not settle the manuscript’s conjectured depth-two formula for S_6 .

12.1 Positive forms and two commuting involutions

Use the five formal letters

$$z = \frac{dt}{t}, \quad c = \frac{dt}{1-t}, \quad a = \frac{i dt}{1-it}, \quad b = -\frac{dt}{1+t}, \quad \bar{a} = -\frac{i dt}{1+it}.$$

Letters stand both for these forms and for a basis of a rational vector space V . Let $H(w)$ denote their outer-letter-first iterated integral from zero to one. An admissible nonempty word has first letter different from c and last letter different from z . Each such integral is absolutely convergent: the last-letter condition controls zero, and the first-letter condition controls one. The other endpoint singularities are logarithmic and integrable.

Define the letter map $\tau = -\phi^*$ by

$$\begin{array}{c|ccccc} v & z & c & a & b & \bar{a} \\ \hline \tau(v) & c-b & z+b & b-\bar{a} & b & b-a. \end{array} \quad (12.1)$$

Put

$$C(v_1 \cdots v_n) = \tau(v_n) \cdots \tau(v_1), \quad K(z) = z, \quad K(c) = c, \quad K(b) = b, \quad K(a) = \bar{a}, \quad K(\bar{a}) = a.$$

Here products inside a word are concatenations; linear combinations are expanded multilinearly. Directly from the table,

$$C^2 = K^2 = 1, \quad CK = KC.$$

Moreover C preserves admissibility. Indeed τ maps the span of $\{z, a, b, \bar{a}\}$ to the span of $\{c, a, b, \bar{a}\}$ and maps the latter span back to the former.

Changing variables and reversing the integration simplex gives

$$H(w) = H(Cw) \tag{12.2}$$

for every admissible word. The sign from reversing the integration path is included in the definition $\tau = -\phi^*$. Thus there is no omitted weight-dependent sign in (12.2). Conjugation gives $H(Kw) = \overline{H(w)}$.

Let $\mathcal{A}_n$ be the rational span of admissible words of length n , and put $\mathcal{A}_0 = \mathbb{Q}$. Superscripts $+$ and $-$ on these spaces will mean the $+1$ and -1 eigenspaces of K . The minus space encodes imaginary parts; choosing one word from each nontrivial conjugate pair is an equivalent coordinate convention.

12.2 All-weight rank of the unmultiplied Cayley rows

Theorem 12.1 (Raw convergent Cayley rank). *For $n \geq 2$,*

$$d_n := \dim_{\mathbb{Q}} \mathcal{A}_n^- = 8 \cdot 5^{n-2} - 2 \cdot 3^{n-2}.$$

The span of the imaginary projections of all weight- n rows $w - Cw$ has rank

$$r_n = \begin{cases} 4 \cdot 5^{n-2} - 3^{n-2}, & n \text{ even}, \\ 4 \cdot 5^{n-2} - 3^{n-2} - 2 \cdot 5^{(n-3)/2}, & n \text{ odd}. \end{cases} \tag{12.3}$$

At $n = 1$, the imaginary space has dimension one and the rank is zero.

Proof. For $n \geq 2$, write

$$U = \text{span}\{z, a, b, \bar{a}\}, \quad E = \text{span}\{c, a, b, \bar{a}\}.$$

Then $\mathcal{A}_n = U \otimes V^{\otimes(n-2)} \otimes E$. The dimensions of U, E, V are 4, 4, 5 and their conjugation traces are 2, 2, 3. Therefore

$$\dim \mathcal{A}_n = 16 \cdot 5^{n-2}, \quad \text{tr}(K \mid \mathcal{A}_n) = 4 \cdot 3^{n-2},$$

which proves the formula for d_n .

It remains to compute traces of reversal. For maps $T : U \rightarrow E$ and $S : E \rightarrow U$, the map $u \otimes e \mapsto Se \otimes Tu$ has trace $\text{tr}(ST)$. Applied to an endpoint pair of C , this trace is four, because $\tau^2 = 1$. Each paired pair of internal letters contributes five. The same facts hold for CK , because $(\tau K)^2 = 1$. Finally the table gives

$$\text{tr}(\tau \mid V) = 1, \quad \text{tr}(\tau K \mid V) = -1.$$

Thus, for $m \geq 1$,

$$\begin{array}{c|cc} n & \text{tr } C & \text{tr } CK \\ \hline 2m & 4 \cdot 5^{m-1} & 4 \cdot 5^{m-1} \\ 2m+1 & 4 \cdot 5^{m-1} & -4 \cdot 5^{m-1}. \end{array}$$

Because C and K commute,

$$\text{tr}(C | \mathcal{A}_n^-) = \frac{1}{2}(\text{tr } C - \text{tr } CK).$$

The image of $1 - C$ is precisely the negative eigenspace of C . Its rank on $\mathcal{A}_n^-$ is consequently $\frac{1}{2}(d_n - \text{tr}(C | \mathcal{A}_n^-))$. Substitution proves (12.3). Weight one is checked directly: K negates $a - \bar{a}$ and C fixes it. $\square$

In particular, the raw weight-seven Cayley matrix has 24 514 imaginary coordinates and rank 12 207. There remain 12 307 coordinates in that particular quotient. This count is exact even though writing and row-reducing the full matrix is unnecessary.

12.3 Closing the relations under shuffle multiplication

The raw row span is not the same as the shuffle ideal generated by the rows. For example, a product of two C -negative elements is C -positive; it vanishes in the ideal quotient but need not belong to the raw negative eigenspace.

Let

$$\mathcal{A} = \bigoplus_{n \geq 0} \mathcal{A}_n$$

with the shuffle product. Both C and K are graded algebra automorphisms: reversal preserves a shuffle, as does applying a linear letter map. Define the homogeneous ideal and quotient

$$\mathcal{J} = \langle f - Cf : f \in \mathcal{A} \rangle_{\text{shuffle}}, \quad \mathcal{B} = \mathcal{A}/\mathcal{J}.$$

Thus every shuffle multiple of every convergent Cayley row is included. The ideal is stable under K .

Let

$$h_n = \dim \mathcal{B}_n, \quad g_n = \text{tr}(K | \mathcal{B}_n), \quad H(t) = \sum_{n \geq 0} h_n t^n, \quad G(t) = \sum_{n \geq 0} g_n t^n.$$

The letter g_n in this subsection is a character trace, not a Gaussian double $g_{a,b}$.

Theorem 12.2 (The full Cayley shuffle-ideal quotient). *The algebra $\mathcal{B}$ is a polynomial algebra on homogeneous generators. Its Hilbert and conjugation-character series are the unique formal power series with constant term one satisfying*

$$\frac{H(t)^2}{H(t^2)} = \frac{1-t}{1-5t}, \quad \frac{G(t)^2}{H(t^2)} = \frac{(1-t)^2}{(1+t)(1-3t)}. \quad (12.4)$$

In particular,

$$\dim \mathcal{B}_n^- = \frac{h_n - g_n}{2}, \quad \dim \mathcal{B}_n^+ = \frac{h_n + g_n}{2}.$$

An explicit product form is

$$H(t) = \prod_{j \geq 0} \left(\frac{1-t^{2^j}}{1-5t^{2^j}} \right)^{1/2^{j+1}}, \quad (12.5)$$

where each root is expanded with constant term one.

Proof. We supply the algebraic details, including the endpoint adjustment. Let $\tilde{\mathcal{A}}$ be the full shuffle algebra on V , including words that are not admissible. This is used only as a formal algebra in the argument.

The shuffle polynomial theorem says that the Lyndon words freely generate $\tilde{\mathcal{A}}$ over $\mathbb{Q}$ [11, 12]. Use the alphabet order $z < a < b < \bar{a} < c$. Every Lyndon word of length at least two is admissible: it cannot begin with the greatest letter c , or end with the least letter z . Conversely, the nonincreasing Lyndon factorization of an admissible word contains neither a one-letter factor c at the beginning nor a one-letter factor z at the end. The usual triangular shuffle factorization therefore gives

$$\tilde{\mathcal{A}} \simeq \mathcal{A}[z, c] \quad (12.6)$$

as graded polynomial algebras. In particular $\mathcal{A}$ itself is polynomial, and its indecomposables agree with those of $\tilde{\mathcal{A}}$ in every degree at least two.

For a connected graded algebra write $Q(\mathcal{A}) = \mathcal{A}_{>0}/(\mathcal{A}_{>0})^2$. Since C and K are commuting involutions over $\mathbb{Q}$, choose a homogeneous simultaneous eigenbasis of $Q(\mathcal{A})$. Averaging arbitrary homogeneous lifts with the four character projectors of the group generated by C, K produces simultaneous eigenvector lifts in $\mathcal{A}$. These lifts freely generate $\mathcal{A}$: replacing a graded polynomial generating set by lifts of another basis of its indecomposables is a degree-triangular invertible change of generators.

For this generating set, $\mathcal{J}$ is exactly the ideal generated by the C -negative generators. One inclusion follows from $x - Cx = 2x$ for each such generator. For the other, after all C -negative generators are set to zero, C acts as the identity on every remaining polynomial. Thus every $f - Cf$ lies in that ideal. Consequently $\mathcal{B}$ is the polynomial algebra on the C -positive generators, retaining their K -parities.

We compute the required generator characters without constructing the generators. The antipode of the full shuffle Hopf algebra is $S(w) = (-1)^{|w|} \text{rev}(w)$, and every connected graded Hopf antipode acts as minus the identity on indecomposables. Hence reversal acts on degree- n indecomposables as $(-1)^{n-1}$. Put $\theta = -\tau$, applied letterwise. On indecomposables of every positive degree we therefore have

$$C = -\theta.$$

In particular, the C -positive generator space is the θ -negative generator space, with the same K -characters.

Temporarily work in the full shuffle algebra. Let $P(t)$ be the Hilbert series of the polynomial algebra on its C -positive generators, and $Q(t)$ the corresponding K -trace series. The letter traces are

$$\frac{\sigma}{\text{tr}(\sigma \mid V)} \mid \begin{array}{cccc} 1 & \theta & K & K\theta \\ 5 & -1 & 3 & 1. \end{array}$$

Since the degree- n word space is $V^{\otimes n}$, the four full-shuffle character series are respectively

$$\frac{1}{1-5t}, \quad \frac{1}{1+t}, \quad \frac{1}{1-3t}, \quad \frac{1}{1-t}.$$

A θ -negative polynomial generator of degree n contributes $(1+t^n)/(1-t^n)$ to the ratio of the identity-character series to the θ -character series. Taking the product over all such generators gives

$$\frac{P(t)^2}{P(t^2)} = \frac{1+t}{1-5t}.$$

If the same generator has K -sign $\varepsilon \in \{1, -1\}$, its contribution to the ratio of the K -character to the $K\theta$ -character is $(1 + \varepsilon t^n)/(1 - \varepsilon t^n)$. Since $\varepsilon^2 = 1$, this gives

$$\frac{Q(t)^2}{P(t^2)} = \frac{1-t}{1-3t}.$$

These character products are valid for an infinite generating set, since only finitely many generators contribute to any fixed weight.

Finally, (12.6) removes two degree-one indecomposables and none in higher degree. On their quotient space, C interchanges z and c modulo $\mathcal{A}_1$, while K is the identity. Exactly one removed generator is C -positive, and it is K -positive. Therefore

$$H(t) = (1-t)P(t), \quad G(t) = (1-t)Q(t).$$

Substituting these two relations proves (12.4). Coefficients are determined recursively because the coefficient of the new h_n or g_n in its square is two. Iterating the first equation proves (12.5) in the t -adic topology. $\square$

The first coefficients and their interpretation are

n	1	2	3	4	5	6	7
$\dim \mathcal{A}_n^-$	1	6	34	182	946	4838	24514
r_n	0	3	15	91	463	2419	12207
h_n	2	9	36	162	720	3312	15336
g_n	0	3	4	18	36	124	300
$\dim \mathcal{B}_n^-$	1	3	16	72	342	1594	7518
$\dim \mathcal{J}_n^-$	0	3	18	110	604	3244	16996

At weight seven, shuffle multiplication supplies 4789 additional imaginary relation directions beyond the raw Cayley row span. The remaining 7518 coordinates are independent only in this specified formal quotient. Adding double shuffle can reduce it further; the dimension of an intersection of relation spaces is not obtained by adding their separate ranks.

12.4 Asymptotic size and the conjugation imbalance

Corollary 12.3 (Two distinct exponential scales). *As $n \rightarrow \infty$,*

$$h_n = \sqrt{\frac{4H(1/25)}{5\pi}} \frac{5^n}{\sqrt{n}} \left(1 + O(n^{-1})\right), \quad (12.7)$$

$$g_n = \sqrt{\frac{H(1/9)}{3\pi}} \frac{3^n}{\sqrt{n}} \left(1 + O(n^{-1})\right), \quad (12.8)$$

$$\dim \mathcal{B}_n^- = \sqrt{\frac{H(1/25)}{5\pi}} \frac{5^n}{\sqrt{n}} \left(1 + O(n^{-1})\right). \quad (12.9)$$

Both of the first two expansions admit all orders in $1/n$. More precisely, the relative difference between the real and imaginary dimensions satisfies

$$\frac{\dim \mathcal{B}_n^+ - \dim \mathcal{B}_n^-}{\dim \mathcal{B}_n^+ + \dim \mathcal{B}_n^-} = \sqrt{\frac{5H(1/9)}{12H(1/25)}} \left(\frac{3}{5}\right)^n \left(1 + O(n^{-1})\right).$$

Proof. The product (12.5) is convergent and nonvanishing in $|t| < 1/5$. The functional equations give

$$H(t) = F(t)(1 - 5t)^{-1/2}, \quad F(t) = \sqrt{(1 - t)H(t^2)},$$

and

$$G(t) = E(t)(1 - 3t)^{-1/2}, \quad E(t) = (1 - t)\sqrt{\frac{H(t^2)}{1 + t}}.$$

The factors F, E are analytic on the disk $|t| < 1/\sqrt{5}$; the roots are the analytic roots taking value one at zero. Consequently the only dominant singularity for H is $1/5$, and that for G is $1/3$. The standard binomial coefficient asymptotic $[t^n](1 - \lambda t)^{-1/2} = \lambda^n (\pi n)^{-1/2} (1 + O(n^{-1}))$, together with Taylor expansion of the analytic factor at the singularity, yields the first two assertions. The factor values are

$$F(1/5) = \sqrt{\frac{4}{5}H(1/25)}, \quad E(1/3) = \sqrt{\frac{1}{3}H(1/9)}.$$

The same Taylor expansion gives every further inverse-power coefficient. Finally $\dim \mathcal{B}_n^\pm = (h_n \pm g_n)/2$ and division proves the ratio. In particular, the 3^n character contribution is not absorbed into an assertion that a finite $5^n/n$ expansion resolves exponentially smaller terms: it has its own independently defined series G . $\square$

12.5 A completely explicit Cayley family containing S_6

Retain $S_p = \sum_{n \geq 0} (-1)^n H_n / (2n + 1)^p$. The harmonic parity split and the positive-form convention give

$$S_p = \operatorname{Im} H(z^{p-1}a(a + \bar{a})).$$

This follows absolutely for $p > 1$, and by a radial Abel limit for $p = 1$.

Proposition 12.4 (An all-index pure-depth representation). *For every integer $p \geq 1$,*

$$S_p = \operatorname{Im} H \left((2ba - aa + a\bar{a})(c - b)^{p-1} \right). \quad (12.10)$$

All products and powers inside H are concatenations. The expanded right side contains exactly $3 \cdot 2^{p-1}$ admissible words, each of weight and series depth $p + 1$. In particular

$$S_6 = \operatorname{Im} H \left((2ba - aa + a\bar{a})(c - b)^5 \right).$$

Proof. Cayley transport gives

$$C(z^{p-1}a(a + \bar{a})) = (2b - a - \bar{a})(b - \bar{a})(c - b)^{p-1}.$$

Both $2b - a - \bar{a}$ and $c - b$ are fixed by conjugation. Taking half the difference with the conjugate and then grouping the three conjugate pairs gives exactly the imaginary part in (12.10). The first letters in the resulting three word families are a or b , and all letters are nonzero kernel letters. Thus every expanded word is admissible and represents an all-one colored multiple polylogarithm. $\square$

This proposition provides a new explicit high-depth expression and a compact formal certificate for the weight-seven mixed sum. It does not turn the manuscript's proposed depth-two S_6 reduction into a theorem. The latter still requires an exact combination of additional relations.

For a real-integral check of conventions, the same identity yields

$$S_p = \frac{1}{(p-1)!} \int_0^1 \log^{p-1} \left(\frac{1+u}{1-u} \right) \frac{\log((1+u)^2/(2(1+u^2)))}{1+u^2} du.$$

Indeed the iterated integral of $(c-b)^{p-1}$ up to u is the displayed logarithmic power divided by $(p-1)!$. Integrating the two outer letters in $2ba - aa + a\bar{a}$, and taking imaginary parts, produces the remaining kernel. The substitution $u = (1-x)/(1+x)$ returns the original Mellin representation of S_p .

There is also a direct verification of this real integral, independent of the word expansion. Start from the already convergent identity

$$S_p = -\frac{1}{(p-1)!} \int_0^1 (-\log x)^{p-1} \frac{\log(1+x^2)}{1+x^2} dx.$$

With $x = (1-u)/(1+u)$ the endpoint pair $(0, 1)$ becomes $(1, 0)$, and

$$-\log x = \log \frac{1+u}{1-u}, \quad \frac{dx}{1+x^2} = -\frac{du}{1+u^2}, \quad \log(1+x^2) = -\log \frac{(1+u)^2}{2(1+u^2)}.$$

Substitution, including the reversed integration limits, gives the displayed formula. Near $u = 1$ the new logarithmic kernel vanishes quadratically in $1-u$, so every fixed logarithmic power remains integrable. At $u = 0$ the integrand is bounded (and has a zero of order $p-1$ when $p > 1$).

12.6 Finite verification and the remaining mixed reduction

The supplied program `check_cayley_ranks.py` is independent of the older report's word implementation. It builds all admissible words, expands the displayed letter transformation, checks $C^2 = 1$ and $CK = KC$, and computes rational ranks by integer elimination with gcd normalization. It also constructs every shuffle lift $H(v)H(w - Cw)$ with total weight at most four.

The exact matrix results are

n	1	2	3	4
$\dim \mathcal{A}_n$	3	16	80	400
raw full rank	1	6	38	190
all lifted full rank	1	7	44	238
raw imaginary rank	0	3	15	91
all lifted imaginary rank	0	3	18	110

The weight-four lifted calculation contains all 1136 prescribed row instances. These are exact rational checks, not modular rank estimates. They audit the all-weight proofs; the proofs do not depend on extrapolating the table.

The separate program `verify_harmonic_cayley.py` rebuilds (12.10) for $1 \leq p \leq 8$ and obtains zero exact residual in every case. The file `S6_cayley_depth_seven.json` lists all 96 terms for $p = 6$. The verifier imports no numerical polylogarithm evaluator.

An exploratory weight-seven double-shuffle computation generated 51 244 normalized rows, in addition to a selected Cayley row. The sparse search was stopped after substantial fill-in; it produced neither an exact certificate for the conjectured S_6 formula nor a separating functional. No membership or nonmembership claim follows from that unfinished computation. The exact quotient and rank theorems above concern the fully specified Cayley-only presentations and remain unaffected.

13 Verification, correction proposals, and integration

13.1 What the computations establish

The proof-bearing arithmetic uses exact integers, rational intervals, or rational polynomial identities. The analytic theorems additionally require the convergence, sign-change, localization, and asymptotic arguments written in the preceding sections. The code is not a substitute for those arguments and is not a proof-assistant formalization.

The included exact replays cover:

1. the two Bernstein positivity certificates, their substitutions, the local angular coefficient, and the rational inequalities in the normalized-radius proof;
2. outward-rounded rational certificates for two fractional thresholds, opposite quartic signs, and the local maximum and minimum scaling prefactors;
3. exact shifted gamma moments and two independent symbolic derivations of the transition's first correction;
4. 48 rational sign enclosures for $f_{n,k}(2)$ and the elementary logarithm bounds used in the spectral-tail estimate;
5. exact Faulhaber generation of Bessel profile operators and simple zero coefficients through four inverse powers of k in the scaled variable;
6. all raw and shuffle-lifted Cayley matrices through weight four, including every one of the 1136 prescribed lifted row instances at weight four;
7. exact zero residuals for the explicit S_p Cayley expression for $1 \leq p \leq 8$, together with the full 96-term expansion at $p = 6$.

The separate diagnostic files contain 24 regularized moment comparisons, 22 endpoint-scale examples, five angular roots, 30 elementary Bessel-zero comparisons, six full-Lerch comparisons, and 12 reflected-moment quadratures. Six of the latter are computed in both original and logarithmic coordinates. The kernel calculations use 80 decimal digits; the Bessel comparisons use 250; the gamma quadratures use 65. The precision and the interpretation of each table are stored with it. These numbers are diagnostic precision settings, not certified error estimates for every displayed decimal.

13.2 The existing conjectures and their current status

The local normalized-radius question is resolved by Theorem 7.2. The global-radius claim for the stated integer range remains open. Theorems 8.1 and 8.2 prove that a blanket fractional extension fails. In particular, a sign in a Taylor expansion near zero does not prove monotonicity on the complete radius interval.

For clarity, one frozen form of the manuscript's S_6 conjecture is

$$S_6 \stackrel{?}{=} \frac{722}{527}g_{6,1} + \frac{40}{527}g_{4,3} - \frac{128}{155}g_{2,5} + \frac{15191}{28569600}\pi^7 \\ - \frac{3}{124}\beta(2)\zeta(5) - \frac{2373}{10540}\beta(4)\zeta(3) - 2\beta(6)\log 2. \quad (13.1)$$

The source already contains a rigorous very small residual enclosure and proves equivalence with another proposed basket. Neither fact establishes equality. Proposition 12.4 gives an exact expression of larger depth, while Theorem 12.2 identifies an entire symmetry quotient. Neither closes the remaining double-shuffle reduction in (13.1).

13.3 Corrections that are justified, and claims that should not be changed

The audit did not find a newly verified false theorem in the canonical portions used here. It did identify stale statements in historical reports, and the extension work exposes several places where a proposed broader statement would be false or unjustified. The integration ledger records these separately.

Stale S_4 status. The pending Lerch report’s attribution paragraph still says that it does not resolve the mixed S_4 conjecture. As a statement about that report’s own contribution this is harmless, but as a current-status description it is obsolete. An editorial annotation should say: “The canonical manuscript now proves S_4 by an exact convergent Cayley and double-shuffle certificate; the mixed S_6 reduction remains conjectural.” Historical source packages can be retained unchanged, with the annotation in their integration record.

Critical endpoint atoms. When extending the signed kernel to $a + b = 1$, the term containing $1/\Gamma(a + b - 1)$ cannot simply be deleted. The family $e^{-T}T^{\varepsilon-1}/\Gamma(\varepsilon)$ tends weakly to an atom at $T = 0$. Omitting it loses mass $(1 - b)^{-1}$ and violates even the zeroth moment. Theorem 3.1 gives the correct measure statement and normalization. This is a correction to the naive extension, not to the earlier theorem for $a \geq 1$.

Boundary Fourier language. In the atomic cases A_n tends to a nonzero constant, so the complex Fourier terms $A_n e^{in\theta}$ do not tend to zero. The correct unit-circle interpretation is the Abel value of the analytic continuation. Every angular theorem also concerns $0 < \theta < \pi$: the real endpoints have zero imaginary part automatically.

Limits of moment and zero statements. The fixed-reflected-exponent leading formula must retain “ m fixed”. Its unrestricted growing- m extension has limiting relative factor $e^{Ce^{-c}}$, which differs from one. Likewise, a fixed-index Stieltjes expansion cannot be used without proof when $n \sim k$. The Bessel theorem resolves one such new regime, and deliberately does not supply a global count of all its positive zeros.

Relation spaces and numerical independence. The raw Cayley image and its shuffle ideal differ already in low weights. In weight seven their imaginary ranks are respectively 12207 and 16996. Neither rank is a dimension theorem for numerical periods, and neither can simply be added to a double-shuffle rank: the intersection must be computed. Similarly, failure of a sparse search to finish is neither nonmembership nor a separating functional.

13.4 Reproduction and repository placement

The proposed destination is `Analysis/Polylogarithms/docs/reports/fractional-cayley-scaling/`. The package is self-contained: `article.tex` inputs the section fragments and bibliography,

and the vector figures are already included. The exact replay command is

```
python3 code/run_verification.py
```

Adding `--diagnostics` repeats the slower numerical checks. The coefficient and matrix programs can also be run separately. The `Makefile` builds the PDF using `latexmk`; `code/build_pdf.py` supplies a simple multi-pass fallback. The machine-readable verification report records exit statuses, software versions, and the scope of each job.

`integration/INTEGRATION.md` maps new results to the existing chapters and distinguishes completed, partially resolved, and open items. `integration/source_snapshot.json` records the pinned revision and hashes of the source files used in the audit. `MANIFEST.sha256` covers the delivered files. The large unfinished S_6 matrices and retrieved historical report copies are excluded; the new reproducible results do not depend on them.

14 Further research questions and proposed programme

The following questions are concrete continuations of the proved results. Each item identifies the missing mathematical step, so that a future finite experiment can be judged against a defined goal.

14.1 Angular geometry and fractional kernels

1. Complete the original global normalized-radius conjecture. Prove or refute that $\eta_{a,b}(\rho)$ decreases on $(0, 1]$ for every integer $a \geq 2, b > 0$, and establish the proposed $a = 1$ trichotomy. Theorem 7.2 removes the first local obstruction. A useful next step is an exact sign formula for $\partial_\rho \eta$ evaluated at its implicit zero, followed by a variation-diminishing or ordered-kernel argument. A proof must control the moving zero, not merely the integrand at a fixed angle.

2. Classify fractional radial turning points. For $0 < b < 1$, the curve $a_*(b)$ determines the local quadratic coefficient. Theorems 8.1 and 8.2 prove opposite local bifurcations on this curve, so the quartic sign is not constant. Determine the number of zeros of $Q(a_*(b), b)$, certify the diagnostic crossing near $b = 0.9974937898734204$, and analyze the sixth-order coefficient there. This is a specific candidate for a higher-order radial transition. Determine also whether a single global radial curve can have several turning points; the local theorems do not exclude that possibility.

3. Determine the complex zero region below the moment boundary. The finite signed-measure classification is sharp, but it does not classify all parameters for complex nonvanishing. For $a = 0$, $F_{0,b}(z) = z \operatorname{Li}_b(z)/(1 - z)$ gives a different route. Determine whether nonzero complex zeros appear for $0 < a < 1 - b$, and, if they do, locate the first parameter boundary. A generalized Stieltjes transform or a distributional moment representation may replace finite variation; it would require a new zero argument.

4. Resolve the two-parameter corner $a \downarrow 0, b \rightarrow 1$. For fixed $b \neq 1$ the crossing approaches one by a power law, while at $b = 1$ its distance is essentially $e^{-1/a}$. Derive a uniform interpolation when $b - 1$ and a vanish together. The exact rescaled kernel is a useful starting point, but neither fixed- b asymptotic is uniform at this corner. Determine the corresponding total-variation profile and the weak limits after renormalization.

5. Optimize the Euler constant in the extended domain. The constants $(1-b)^{-1}$, $\zeta(b)$ and $e^{a(\gamma+\psi(a))}/(a^2\Gamma(a))$ give explicit enclosures. They are not claimed to be optimal remainder constants. Characterize the exact signed mass as a function of (a, b) , sharpen the remainder using its vanishing moments, and design an interval evaluator that remains effective as the positive density collapses into an atom.

14.2 Lerch zeros and overlapping asymptotic regimes

6. Unfold the central Bessel zero. For fixed $r > 0$, Ψ_r has a multiple zero at zero. Its splitting depends on a spectral contribution that is invisible to every algebraic order in $1/k$. Obtain its leading exponential scale and sign, including possible cancellation among shifted terms. This should connect the fixed- (n, k) small- ρ Puiseux theorem with the joint limit $n = k + r$, but the limits need not commute.

7. Allow the Bessel order or zero label to grow. Determine a range of $r = r(k)$ and $m = m(k)$ in which the fixed-order Bessel approximation is still uniform. Find the transition to an appropriate large-order or turning-point profile. The first technical obligation is a uniform coefficient bound for the symmetric product after the factorial transform, rather than a fixed- r formal expansion.

8. Sharpen the exponential Lerch deformation. The factor $e^{-k/20}$ is explicit and adequate for separation of scales, but it is deliberately conservative. Optimize the coefficient contour and derive the actual leading exponential term, including its oscillatory sign. This would explain the changes of sign seen in the full-Lerch diagnostics and quantify when the first shift dominates.

9. Determine the sign pattern of $f_{n,k}(2)$. The all-pair small- ρ theorem reduces the remaining count choice to a nonzero integer polynomial evaluated at $\log 2$. Find a useful closed sign criterion in large regions of the (n, k) lattice, or an asymptotic sign law when both indices grow. Transcendence proves nonvanishing, but supplies no practical separation bound. Rational interval certificates can produce a reliable initial map.

14.3 Reflected moments and uniform asymptotics

10. Extend the critical transition to complete finite orders. The gamma localization argument and exact moments give an algorithm for higher terms. Compute and prove a useful recurrence for the coefficient functions beyond P_1 , determine whether all negative powers of L cancel, and produce explicit computable remainder constants on prescribed compact c intervals. The first cancellation is proved here; a general cancellation pattern needs its own argument.

11. Treat proportional growth and symmetry of the saddle. For $m/n \rightarrow \lambda > 0$, analyze the critical points of

$$\log \log \Gamma(x) + \lambda \log \log \Gamma(1-x).$$

Prove uniqueness or classify competing maxima, obtain a uniform Laplace expansion, and connect it to the $m \asymp n/\log n$ boundary regime. The exact symmetry $M_{n,m} = M_{m,n}$ is an essential consistency test. Numerical maximization alone cannot exclude an additional saddle.

12. Make optimal truncation uniform in the reflected exponent. The source’s sharp fixed- m remainder is a separate theorem from the present joint transition. Determine a common range in (n, m, N) for which late inverse-function coefficients and the least-term remainder are uniform. The early-pole resummation suggests the correct matching, but does not license summation of a divergent infinite pole series.

14.4 Formal relation quotients and special values

13. Prove the frozen S_6 depth-two formula. Construct simultaneous Cayley and conjugation eigenvector generators, reduce the weight-seven problem to the 7518-dimensional imaginary Cayley quotient, and impose double shuffle there with exact lifting data. A successful output is a rational certificate whose expanded residual is exactly zero. A negative answer for a specified presentation requires a separating functional; it does not disprove the numerical identity unless the presentation is known to be complete.

14. Determine the intersection with double shuffle. Theorem 12.2 answers the Cayley-only problem in all weights. Find the Hilbert or character series after adding a precisely specified regularized double-shuffle ideal. Compare raw rows, multiplicative closures, distribution, and rational pullback relations. Any dimension conjecture should specify which algebra and which ideal are meant, and should distinguish formal dimensions from periods.

15. Generalize the involution quotient theorem. The proof uses a graded polynomial shuffle algebra, two commuting involutions, and a finite endpoint correction. Extend it to other root-of-unity alphabets and rational changes of variables when the admissible subalgebra is stable. Determine which transformations can be simultaneously linearized on generators and what replaces the dyadic functional equation when the acting symmetry has higher order.

16. Search for short consequences of the explicit S_p family. The all-depth expression is exact but expands exponentially. Seek families in which conjugation, parity, or shuffle symmetries collapse most terms, and identify weights where a depth-two basket can plausibly be complete. Freeze each candidate basket and coefficient vector before verification, retain exact residual enclosures, and then seek symbolic proofs. The rejected earlier S_8 vector should remain a regression case rather than being silently replaced.

14.5 Suggested order of work

The most immediate finite task is the equivariant Cayley reduction for S_6 , because the quotient theorem gives a concrete smaller target. On the analytic side, the global integer-order radial conjecture and the two-parameter kernel corner are natural next goals: the present sign and scaling formulas expose their remaining obstacles. The central Bessel unfolding and uniform late-term gamma theory are more demanding matching problems. All four directions benefit from keeping exact finite verification separate from conjectural global extrapolation.

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